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**Personnel/human resource departments and uncertainty: A test
of Thompson's model of boundary spanning units**

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Georgia Institute of Technology, 1989

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**PERSONNEL/HUMAN RESOURCE DEPARTMENTS
AND UNCERTAINTY: A TEST OF THOMPSON'S MODEL OF
BOUNDARY SPANNING UNITS**

A THESIS
Presented to the Academic Faculty

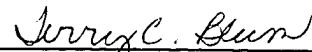
By
Nathan Bennett

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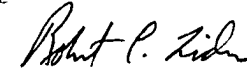
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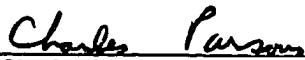
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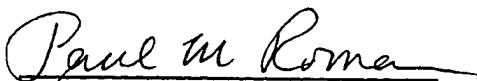
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SUMMARY

Efforts to understand how an organization monitors its environment often focus on boundary spanning individuals or units. The concept of boundary spanning was one introduced by Thompson in 1967 in a book that has since become the classic statement of the open systems model of organizations. In the twenty years since its publication, considerable research on some portions of Thompson's theory has been conducted. This research has often focused on the psychological consequences faced by individuals who engage in boundary spanning activity. After reaching a peak in the 1970's however, this line of research has all but disappeared from the journals.

It is suggested herein that there have been three general problems with the research to date testing Thompson's concept of boundary spanning. First, investigators have focused too narrowly on boundary spanning activity, ignoring the more complex nature of the boundary spanning role. Second, notable by its absence is a body of research which focuses on what Thompson called boundary spanning units as opposed to individuals with boundary spanning activity. Third, the role of the boundary spanner as an uncertainty absorbing mechanism is a major proposition in Thompson's model that has gone almost entirely untested. The author is aware of only one study (Daft, Sormunen, and Parks, 1988) that has examined boundary spanning and its relationship to perceptions of environmental uncertainty.

This dissertation uses Thompson's model of boundary spanning units to generate hypotheses which are then tested in a sample of personnel/human resource departments. A theory driven argument is presented which suggests that the P/HR function in an organization meets Thompson's criteria for a boundary spanning unit. First, an historical

review of the roots of personnel administration suggests that its emergence as a field fits Thompson's conceptualization of the development of boundary spanning units. Second, common themes in current personnel/human resource management literature concerning the "upgrading" of the field from a relatively low level, administrative function to one that is an integral part of an organization's functioning are examined. While it is reasonable to suspect that the new language in P/HR literature reflects an actual change in P/HR practice, there is very little evidence to empirically document this transition. For the most part, literature on what is now known as "strategic human resource management" is prescriptive, vaunting the numerous advantages hypothetically possessed by companies that take such an approach to human resources. The dissertation suggests that from its origin through this most recent period of transition, the development of the personnel function (now generally referred to as human resource management) has been due to a recognition by organizations that an increasing complexity and concomitant uncertainty in the human resource arena caused by such factors as labor unions, the expansion of due process in organizations, and intervention by government agencies, needed to be addressed.

The dissertation tests hypotheses generated from Thompson's theory of boundary spanning units using a sample of personnel/human resource departments ($N = 124$) in the eighteen county Atlanta Standard Metropolitan Statistical Area. The data analyzed are from face to face interviews and mailback questionnaires completed by the highest level personnel/human resource person at each site. The results of the analysis support the first hypothesis, which concerns the prevalence of uncertainty in areas nearest the organization's core technology. The second hypothesis describes the relationship between external linkages maintained by the personnel/human resource department and uncertainty and cannot be tested directly, but is examined. The relationship of the two stages of boundary spanning - external and internal linkages - are considered in the third hypothesis. The

fourth hypothesis, which examines the relationship between uncertainty, boundary spanning, and resources committed by the organization to address uncertainty is supported. The results are discussed with regard to both the hypotheses that did and did not receive support, as well as future research.

CHAPTER I

INTRODUCTION

The Organization and its Environment

In order to explain the significance of Thompson's boundary spanning concept, it is important to understand how the role of the environment came to be important to organizational theorists. Certainly an understanding of what constitutes an organization's environment is essential to a theory in which the organization crosses its boundary and enters into interaction with that environment. From the late 1950's through the early 1970's, a number of open systems theorists developed models designed to define and incorporate an organizations environment. Parsons (1958a, b) elaborates on a theory of organizations as open systems. The open systems view represents a drastic departure from the closed system, or "information tight" view of organizations (Ashby, 1958) and has become the dominant paradigm in organizational theory. An open systems perspective recognizes the dependence of the organization on and the existence of uncertainty in the environment. Such a dependency relationship has consequences for many aspects of organizational behavior. Despite the widely recognized importance of the environment in open systems theory, its conceptualization varies.

The work of Emery and Trist (1965) stressed that not only were environments difficult to identify, but they were in a state of flux, becoming more and more complex. They identified four types of environments based on what they conceptualized as the environment's "causal texture". The simplest environment which they labelled placid-randomized, is characterized by a lack of distinction on the part of the organization between strategy and tactics. The second type, placid-clustered, describes an environment of

imperfect competition between organizations. The third type, disturbed-reactive, consists of an oligopolistic market, where more than one organization of each type seeks to compete with and learn about the others. In the most complex type, turbulent fields, impact is felt not only from the components of the environment, but from the environment itself. Emery and Trist argue that the trend is such that organizations will find themselves in turbulent fields, facing exponential increases in the amount of "relevant uncertainty" with which they must cope (1965: 26).

Duncan (1972) identified two dimensions which could be used to conceptualize complexity in the environment. One dimension, called "simple-complex", represents the number of factors considered in decision making by the organization. The second dimension, labelled "static-dynamic", is concerned with the degree to which factors in decision making are the same or different over time. Organizations which find themselves in complex, dynamic environments are faced with greater demands from their environment than are those organizations in simple, static environments.

Pfeffer and Salancik (1978) and Churchman (1968) have suggested that a more realistic and meaningful consideration of the environment treats it as beginning at the point where the organization begins to lose control over activities to other organizations or individuals. Pfeffer and Salancik (1978) have identified three "environments" that could be of concern to organizations. The first is the overall environment, which consists of the entire system in which the organization exists. The second, a subset of the first, is that portion with which the organization is in direct contact. The third type, also a subset of the first, is what Weick (1969) termed the enacted environment. This is the part of the environment that the organization perceives. In terms of organizational theory, it would appear that the second conceptualization of environment is most easily identified, while the third type is more reflective of what the organization considers to be its "reality". The third

type is less easily quantified. Given that organizations are exposed to a great deal of information and that they might find it to be of widely varying importance or relevance to their operation, the means by which organizations assess the utility of information becomes of concern.

In one of the early articles addressing organizational environments, Dill (1962) discussed the difficulties inherent in identifying quantifiable aspects of the environment. Dill noted that this difficulty stems partly from our inability to gain control over the environment, and partly from differences in the way individuals perceive it. Further, it is difficult to know how the organizational members under study conceptualize it (Child, 1972; Pfeffer and Salancik, 1978). In order to identify a manageable, relevant portion of the environment, Dill suggested that it be treated as "information which becomes available to the organization or to which the organization, via search activity, may get access" (1962; 96). Such a conceptualization allows us to consider information that is relevant to the core technology of the organization, and provides clues as to the task environment of that organization. Further, Dill (1962) proposed that information can have a variety of consequences on an organization's behavior. Information can serve as a "trigger" to action, it can provide clues about goals, means, or constraints faced by the organization, and can provide clues relevant to evaluation of the organization's performance. This conceptualization is similar to that of Weick (1969) and Duncan (1972), who were concerned with the flow of information through organizational members.

There are two components present in each of the above conceptualizations of environment that are important to Thompson's boundary spanning concept. The first is that environments are difficult to identify and manage, and are becoming more complex. The second is that the tool used by organizations to try and cope with the environment is information. Thompson's theory suggests that the relations between an organization and

its environment are directed by boundary personnel, who control the flow of information. Terreberry (1968) proposes that an organization's ability to adapt to the continually changing environment will be based on the amount of flexibility in its structure, the ability of the organization to anticipate external demands for change, the quality of the search by the organization regarding efficient use of inputs and outputs, and the quality of the organization's internal processing system. Organizations which strive to develop these abilities often do so through the establishment of boundary spanning units.

Organizational Boundary Spanners

When considering the environment, organizations try to identify homogeneous segments and establish units to deal with each segment (Thompson, 1967). In this way, the organization creates a structure that matches that of its task environment. A central tenet of Thompson's theory is that organizations try to operate under norms of rationality. In other words, organizations wish to make decisions and take action based on complete understanding of all relevant variables. While this is a difficult and likely impossible task, it remains a goal organizations are said to work to achieve. Thompson suggests that when the organization faces a stable and known task environment, it can rely on organizational rules to guide the behavior of its unit members. For example, when organizational members are faced with a certain situation "A", organizational procedures prescribe the proper course of response "B" (Blau, 1955). In practice, not all organizations are faced with such certainty in the environment. When the environment is unpredictable, the organization monitors the environment through specially constructed, localized, boundary spanning units (Thompson, 1967). Boundary spanners are a mechanism that organizations use to help create a situation wherein they can operate under conditions approximating norms of rationality. As Thompson states, "The crucial problem for boundary spanning

units of an organization, therefore, is not coordination (of variables under control) but *adjustment* to constraints and contingencies not controlled by the organization - to what the economist calls exogenous variables" (p 66-67, original emphasis). As will be discussed later, these adjustments can be proactive through efforts to influence the environment, as well as reactive to signals emanating from it.

Boundary spanning theory takes as a given the existence of organization/environment boundaries, which individuals are consequently charged to bridge. On the most basic level, boundary spanners interact with both members and non-members of the organization. To summarize the argument elaborated above, (1) the environment can be conceptualized as the body of information available to the organization (Dill, 1962); (2) organizations need to gather information in order to assess the amount and source of support for their goals (Thompson and McEwen, 1958; Katz and Kahn, 1966); and (3) viable organizations survive based on their ability to learn and perform according to changes in the environment (Terreberry, 1968; Dollinger, 1984). Effective boundary spanners can provide the organization with the information it needs to make effective changes (Hage and Aiken, 1970) thereby enhancing the likelihood of organizational survival (Pruden, 1969; Allen and Cohen, 1969; Organ and Greene, 1972).

Boundary Spanning Roles

Research has shown that boundary spanning is a two-step process, involving both intake and representation functions (Miles, 1976, 1977; Tushman and Scanlan, 1981a). First, boundary spanners scan the environment in order to seek, filter, and select relevant information to be passed on to organizational decision makers. Second, the boundary spanner serves an external representation function; buffering and moderating the effects of, and in some cases influencing, the environment (Aldrich and Herker, 1977). Clearly, this represents a reverse flow of information from the organization core out to the environment

(e.g. Aldrich and Reiss, 1971). In both instances, the commodity in which the boundary spanner deals is information.

One feature of the boundary spanning role, and one that distinguishes it from others with external linkages (e.g. Tushman and Scanlan, 1981a) is that its occupants potentially have a considerable amount of autonomy (Aldrich, 1979). In situations where the environment is predictable and rules can be established to guide boundary interactions, the amount of autonomy is greatly reduced. Generally, however, boundary spanners are faced with vague and often contradictory information which they must evaluate and pass on to others, usually in a simplified form. This process, called "uncertainty absorption" by March and Simon (1958: 165) comes with a great deal of autonomy and requires the intelligent and experienced use of discretion by the role occupant. Theoretically, this autonomy, coupled with continued success in the boundary role can lead to increased power (Pettigrew, 1972). One negative aspect of the power in boundary roles is that it raises questions concerning the loyalty of the boundary role holder to the organization (Organ, 1971a). To control the boundary spanner, organizations can rely on the professionalism and concomitant ethical standards of the boundary role incumbent as a monitoring device, or can give them high positions with the hope of creating greater organizational commitment (Aldrich and Herker, 1977).

A second characteristic of boundary spanning roles is that they are formally designated (Aldrich, 1979). This feature of boundary roles is interesting, because such a designation becomes an unobtrusive indicator of the environmental contingencies which are of concern to the organization. In other words, areas which are perceived as important, i.e. the organization's enacted environment (Weick, 1969), can be identified by the presence of formally designated boundary spanning individuals (Aldrich, 1979). For example, this suggests that an unobtrusive measure of an organization particularly

concerned about legal issues would be the presence of an unusually large staff of corporate lawyers.

To this point, this paper has identified several distinguishing components of boundary spanning based on the theoretical work of Thompson (1967) and others. The boundary spanning individual is designated as such by the organization (Aldrich, 1979) in response to turbulence (Emery and Trist, 1965; Terreberry, 1968) or uncertainty (Thompson, 1967) in the environment. Boundary spanners engage in uncertainty absorption (March and Simon, 1958), which requires autonomy (Aldrich, 1979), and control information which results in power (Pettigrew, 1972; Tushman and Romanelli, 1983). The boundary spanning individual is well connected both within and outside the organization (Tushman and Scanlan, 1981a). These features of this role facilitate the accomplishment of the scanning, buffering, moderating, adjusting and influencing functions in which boundary spanners engage (Thompson, 1967; Aldrich and Herker, 1977). This definition is important to remember when considering the research reviewed below. The review will point to a major problem in tests of Thompson's theory: the confounding of the boundary spanning role just described with that of individuals who display what Tushman and Scanlan (1981a, b) call external linkages and engage in boundary spanning activity.

Research on Boundary Spanning

Broadly defined, boundary spanning activity is done by anyone who interacts with the organizational environment. Using this definition, boundary spanning activity would include any activity involving the interaction of an organizational member with a non member, or exposure to an object, such as a newspaper. Any employee that leaves the worksite is likely to meet this criterion. In this context, "boundary spanning" becomes an extremely diluted theoretical construct. Boundary spanning activity is just another activity

in which many employees engage. Clearly, this is not what Thompson (1967) had in mind when he was concerned with how organizations handle uncertainty in the environment. The most crucial aspect of what Thompson called boundary spanning is that the role is created or recognized by the organization as a way to reduce uncertainty in the environment. This aspect of the role is equally crucial to an open systems theory of organizations. Boundary spanning activity by an employee does not necessarily lead to a reduction in uncertainty for the organization, nor does it indicate anything about the organization's structure in response to the environment. Leifer and Delbecq (1978) have suggested that boundary spanning activity is a more appropriate focus for research than boundary roles, because many activities, such as reading trade magazines, involve interaction with the environment. Certainly, most employees engage in boundary spanning activity when it is defined as any interaction with the environment. The questions this proposal raises are to what extent is this sort of boundary spanning activity relevant to the organization, and to what extent does this reduce or absorb uncertainty for the organization.

Based on the theoretical development of the boundary spanning role described above, a distinction needs to be made between boundary spanning activity and boundary spanners. Boundary spanning activity is activity directed at elements of the environment. It is analogous to what Tushman and Scanlan (1981a, b) call external linkages. Boundary spanning is one activity in which boundary spanners engage in performing their uncertainty absorption role, which has been elaborated earlier. Unfortunately, the fact that most research has used boundary spanning activity to test hypotheses about boundary spanners makes it less relevant to this study. At the same time, a brief review and discussion of the merits and limitations of the research provides background.

Research on boundary spanning individuals is most often concerned with their personal characteristics and implications of that activity. One line of thought suggests that

boundary spanners face a great deal of conflict and ambiguity in their work, due largely to the psychological and structural location of the role "between" organizational and external constituents (Belasco, 1966; Organ, 1971a). The lack of formal authority over external agents hinders the ability of the boundary spanner to control his/her relationships with external agents. As a result, the boundary spanner has to resort to various ingratiating techniques to gain influence, often leading organizational members to be confused concerning the boundary spanner's true loyalties. Further, the boundary spanner is often in a position where he/she must share unpopular information with the organization. Boundary spanners with inadequate status may fear a "kill the messenger" phenomenon. As a result of these features of the boundary role, some authors (e.g. Kahn, Wolfe, Quinn, Snoek, and Rosenthal, 1964) suggest that the boundary role incumbent will experience great levels of stress and discomfort. Conversely, others (e.g. Aldrich, 1979) think that the autonomy inherent in the position leads to greater satisfaction among boundary spanners. Pettigrew (1972) wrote that those who control information hold power. Certainly boundary role incumbents possess a great deal of power within the organization based on their knowledge of the environment. This power could reduce the level of stress felt by boundary role incumbents. While Organ (1971a) states that boundary personnel do not have much power over their external contacts, it would appear that their knowledge of the company's position or the support of their superiors would indeed provide them with leverage in interactions and thereby reduce stress. As could be expected, these theoretical contradictions manifest themselves in the research.

One stream of research has focused on samples of individuals who the various authors argue display highly visible boundary activity. A typical example of the jobs studied is the salesperson. This selection may have occurred for a number of reasons, although the early work of Belasco (1966), using salesmen, probably had an impact on

this. Further, the assumption that salespeople engage in boundary spanning activity has face validity. Research on salesmen has found that high performing salesmen tend to identify more closely with their customers, reporting both business and personal relationships with them (Pruden and Reese, 1972). Pruden and Reese (1972) suggest that salesmen try to develop role sets that allow them to minimize conflict and maximize satisfaction. Spekman (1979) looked at the basis of power of boundary spanners in a sample of 322 buyers for industrial and consumer goods firms. Spekman found that as the constituent's needs for information increased under conditions of high perceived uncertainty, the boundary spanner obtained more power, supporting the relationship discussed by Pettigrew (1972). Dubinsky, Hartley, and Yammarino (1985) used a sample of insurance and retail sales personnel, who they claim are "boundary spanners by definition" (p 289). Concerned with the role of self-monitoring in performance and role conflict, the authors found that holding experience constant, the ability to react to diverse situations was not related to conflict or performance. A weak, positive, relationship was found between self-monitoring and role ambiguity.

Parkington and Schneider (1979) look at relationship between stress and boundary spanning activity. Using a sample of 263 bank employees, they found support for their hypothesis that the relationship between stress and boundary spanning is affected by the amount of discordance between management directives and customer demands. Schneider and Bowen (1985) also considered a sample of bank employees and found some support for the hypothesis that boundary role employees experienced high levels of stress. They also concluded that the banking industry is such that the boundary between organization and customer is blurred. That is, a bank is an open system with highly permeable boundaries. One implication from these studies for the boundary spanning literature is that such poorly defined boundaries may have unique characteristics. For example, the concept

of boundary spanner may lose meaning in a situation where boundaries are so easily crossed in both directions.

The failure of this group of studies as far as the present analysis is concerned is in the focus, noted above, on boundary spanning activity. Bank tellers, for example, interact with the environment of a bank manifested as clients. Salespeople interact with clients who are a very important part of an organization's environment. The previously defined indicators of boundary spanning show that neither the bank teller nor the salesman meet those criteria. Bank tellers are not put into the position because of uncertainty in the environment; they are placed there because you cannot run a bank without them (although technological developments may soon change this, too.). Although it would be wise to test the following assumptions, it seems safe to say that bank tellers do not operate with a great deal of autonomy, they do not wield power within the organization, and they are not seen by organizational members as "experts" who protect the technical core of the organization from environmental influences. These employees coordinate the "variables under control" as opposed to those which are unpredictable (Thompson, 1967). Similar statements could be made and tested about retail salespersons.

A second stream of research has measured boundary spanning activity as a variable (as opposed to a given). Keller and Holland (1975) studied 51 professional employees in a large government research and development organization. Building on the research of Kahn et al (1964), and Organ (1971a, b), the authors hypothesized that boundary spanning activity would be positively related to role conflict and ambiguity and negatively related to job satisfaction. Boundary spanning activity was measured using four items which asked how many publications they regularly read, the frequency with which they recommended information sources to and sought out information sources from a fellow employee and the frequency from which others had sought information from them. These items were

summed and were validated by supervisor's identification of individuals thought to be high on boundary spanning activity. Contrary to their hypotheses but in line with what Aldrich (1979) would have expected, boundary spanning activity was negatively correlated with role ambiguity and positively related to job satisfaction.

Keller, Szilagy, and Holland, (1976) studied 192 managers, engineers, and supervisors to examine the relationships between boundary spanning activity, role conflict, ambiguity and satisfaction. Using the boundary spanning activity measure discussed above, Keller *et al* found that boundary spanning activity was related to job variety, autonomy, task identity, feedback, friendship, friendship opportunities, and dealings with others. At the same time, the relationship between boundary spanning activity, role conflict, and role ambiguity proposed by Kahn *et al* (1964) was not observed. At the lower levels of the organization, however, a tendency towards support of the Kahn *et al* hypothesis was noted. Boundary spanning was significantly related to level in the organization, with those higher up exhibiting more boundary spanning activity. Keller (1978) used the same boundary spanning measure on 132 managerial, engineering, and supervisory employees to look longitudinally at the same set of variables. Cross lagged correlation found there were positive relationships between boundary spanning activity and satisfaction. These findings are interpreted as supporting the opinion offered by Pettigrew (1972). Keller notes that the measure of boundary spanning activity used may tap more into the information transfer aspect of boundary spanning, and miss the external representation aspect which may be more stressful. It seems that Keller's concern is justified. If the boundary spanning measure is examined, it is clear that it measures frequency of interaction with members and non members. Also, very little is learned about the context of the interactions. The measure does not indicate (1) how much the organization depends on the individual's information gathering (Thompson, 1967); (2)

how well connected the individual is within his/her organization (Tushman and Scanlan, 1981b); (3) how much power or autonomy the individual has (Pettigrew, 1972; Aldrich, 1979); and (4) if the organization established the role to deal with perceived uncertainty in the environment (Aldrich, 1979; Thompson, 1967).

Miles (1976) considered boundary spanning activity as one role requirement that could be a source of organizational stress. Boundary spanning activity was assessed through the use of 11 items which addressed how often the individual represented the organization to outsiders, travelled, coordinated the activities of others, gave briefings, and acted as liaison with other organizational units. Contrary to the findings in the work by Keller and his colleagues, the analysis found that role conflict was strongly related to the extent of boundary spanning activity.

The measure of boundary spanning activity used in the Miles studies (1976, 1977) represents an improvement from those used in the Keller studies (Keller and Holland, 1975; Keller *et al.*, 1976) and by others (e.g. Hambrick, 1981). For example, the measure addresses the external representation function which previous measures had overlooked. When considered in light of the definition of boundary roles outlined above, however, it becomes clear that the Miles measure concentrates almost exclusively on the frequency with which information is gathered and transferred. The items used to measure boundary spanning ignore the question of the "connectedness" of the boundary spanner, and the amount of autonomy or power held by the boundary spanner. By excluding these organizational aspects of boundary roles, the studies fail to take into consideration the dependence the organization may have on the boundary spanner, and the influence he/she may have as a result. These are crucial variables which should be incorporated into a model that examines role conflict and ambiguity (e.g. Parasuraman and Alutto, 1984).

Tushman (1977) studied 345 professionals in a research and development laboratory. Boundary spanning was measured by the median amount of communication in a fifteen day period between the department and three external departments. This measure again concentrates solely on the frequency of external contact through boundary spanning activities, at the expense of other crucial boundary spanning role characteristics. The results indicated that subunits faced with greater uncertainty had a greater number of boundary roles. There seemed to be a curvilinear relationship between the number of boundary roles and project performance. In other words, at a point additional boundary roles did not provide increases in performance. The results indicate that "work characteristics do not cause boundary roles; rather, the high-performing work areas develop the appropriate number of boundary roles to fit the external information-processing demands of their work." (p 601).

Tushman and Scanlan (1981a, b) looked at the characteristics of boundary spanners and the extent to which they linked more than one external area to the organization. Of great importance is the authors' distinction between boundary spanning activity (external linkages) and boundary spanning individuals (external and internal linkages). Essentially, those with boundary spanning activity have not shown that their information is disseminated within the organization (Von Hippel, 1976). Boundary spanners, however, are strongly linked inside and outside of the organization (Tushman and Scanlan, 1981b). These two types are to be distinguished as well from "internal stars", a term Tushman and Scanlan use to describe those individuals who are seen by their colleagues as technically competent and are hence frequently consulted. It is precisely these distinctions that previous research had ignored. Here, Tushman and Scanlan found that boundary spanners had specialized characteristics to match the type of environment they were working in (e.g. Kelly and Kranzberg, 1975). They present a model of boundary spanning roles, which

suggests that boundary spanners emerge based on perceived levels of competence in special areas. This is an interesting proposition, and is in contradiction with the literature that suggests organizations designate individuals to fill boundary roles in areas where the environment is perceived as turbulent (Thompson, 1967; Aldrich, 1979). In their model, boundary spanning becomes something one tries to do to enhance their value and gain a promotion, as opposed to the distinguishing feature of the job.

Daft et al (1988) have conducted the one known study which discusses the relationship between boundary spanning and perceptions of uncertainty. In their paper, Daft et al measured environmental scanning (a boundary spanning activity) as a mechanism used by Chief Executive Officers to reduce uncertainty. In the analysis, the researchers found that greater "strategic uncertainty", a concept which integrates uncertainty and importance into one measure, existed in environmental sectors that were closer to the organization's technical core. In instances of high strategic uncertainty, CEOs reported more frequent scanning behavior, and the CEOs in high performing firms engaged in more scanning to reduce uncertainty than CEOs in low performing firms.

Summary and Implications for Present Investigation

In sum, research on boundary spanners has resulted in contradictory findings and has failed to provide an adequate test of Thompson's theory. Generally speaking, studies have operationalized boundary spanning activity, or focused on samples which display high levels of boundary activity, each of which has been shown to be inappropriate for a test of Thompson's theory. Further, propositions concerning the structure and consequences of boundary roles for individuals and organizations cannot be adequately tested on samples that exhibit merely boundary spanning activity. Finally, only one study has examined the relationship of boundary spanning to perceptions of uncertainty in the environment, the most crucial link in Thompson's model.

This last omission from the literature is the most troubling and provides a stimulus for the present investigation. There would appear to be at least two reasons which could explain the lack of empirical examination of this central tenet of Thompson's theory. One explanation is that the assertion that organizations would take action to reduce uncertainty in their environment is so obvious as to need no confirmation. It would appear that there are enough examples in the field where such once obvious relationships have not been supported to eliminate, or at least make implausible, this explanation. These include, for example, the relationship between stress and performance (e.g. Jamal, 1984; 1985) or job satisfaction and performance (e.g. Iaffaldano and Muchinsky, 1985).

A second, more plausible explanation, has both theoretical and methodological bearing for this analysis. In general terms, Thompson's model theorizes that an organization develops boundary spanning units as a reaction to (or in anticipation of) an uncertainty in the environment. At a point in time, the boundary unit enters into interaction with the environment to ascertain the extent of uncertainty. There are several possible outcomes of this interaction. First, the boundary spanner may report that there is "no problem", the original concern was unfounded. Second, a report could be made that "big trouble" lies ahead. In the former case, boundary spanning may cease or may continue in a passive, monitoring mode. In the latter case, boundary spanning continues. If a researcher were to measure the reciprocal relationship between boundary spanning and uncertainty at this point in time, they would find that perceptions of uncertainty lead to greater levels of boundary spanning which in turn leads to greater perceptions of uncertainty. This would appear to contradict Thompson's theory which suggests boundary spanning is an uncertainty reduction mechanism. A more appropriate test of the theory, however, would return to the organization after a period of time when the boundary spanning unit would have either been able to pass information on to decision makers who would develop plans

to address the uncertainty or influence the environment in such a way as to reduce uncertainty. In other words, the organization may have institutionalized strategies to address the environment (e.g. Tolbert and Zucker, 1983). At this point in time, boundary spanning should not be related to higher perceptions of uncertainty. A third possible outcome is that the boundary spanner finds that the organization will not be able to influence the environment or otherwise address the uncertainty with finality. In this case, a steady state of uncertainty exists, and continued monitoring on the part of the boundary spanner would be expected. This statement of the theory is obviously difficult to test from a methodological standpoint. In this analysis, environmental sectors will be identified where it is argued that relationships have stabilized and others that have not yet had time to stabilize in an effort to approximate the design suggested above. Specifically, sectors of the personnel/human resource environment are used to examine the boundary spanning - perceived uncertainty relationship.

CHAPTER II

P/HR MANAGEMENT AS A BOUNDARY SPANNING MECHANISM

The Unit of Analysis in Boundary Spanning Research

In his book, Thompson (1967) introduces the boundary spanning concept through a discussion of boundary spanning units. Thompson states that "organizations subject to rationality norms seek to isolate their technical cores from environmental influences by establishing boundary-spanning units to buffer or level environmental fluctuations" (1967: 67). In other words, the creation of a boundary spanning unit is a conscious, structural change initiated by the organization in an effort to assure an adequate understanding of, and to potentially gain some mastery over, the environment. Thompson's concept of boundary spanning jobs is introduced later and while it is not explicitly stated seem to be those jobs that occur in a boundary spanning unit. The characteristics of boundary spanning noted earlier apply at both the department and job levels.

The important difference between a focus on boundary spanning units and jobs is a level of analysis issue and has bearing on the choice of dependent variables. For example, research on boundary spanning jobs (or roles), explicitly involves an individual level analysis and would be appropriately focused on outcomes the individual faces, such as role conflict or ambiguity. This set of questions has been partially addressed by research on boundary spanning activity. On the other hand, research on boundary spanning units requires an organizational level of analysis, and would more appropriately examine the relationship between boundary spanning and perceptions of environmental uncertainty held by both top management and boundary spanners, and ultimately organizational performance in light of its understanding of the environment. The above review of

boundary spanning research does not contain a single study that examines boundary spanning units. Additionally, with the exception of a recent study by Daft *et al.* (1988), not one study reviewed has examined the relationship between boundary spanning and perceptions of uncertainty. This void is remarkable, given that the theory argues that the role of boundary units is specifically to address environmental uncertainty. Certainly, the observations concerning the increasing environmental complexity and interdependence of organizations made two decades ago by Emery and Trist (1965), Terreberry (1968), and others remain true. Organizations will undoubtedly continue to seek out information in efforts to reduce the uncertainty with which they are continually faced. As the main tool organizations have in seeking out, filtering, and disseminating information, boundary spanning units will continue to be important. In this dissertation, the personnel/human resource function in organizations is considered using the boundary spanning framework.

A Brief History of Personnel Management

A common theme in current personnel/human resource management (hereafter P/HR) literature concerns the "upgrading" of the field from a relatively low level, administrative function to an integral part of an organization's functioning. The transition is certainly reflected in new P/HR texts, which bear such titles as "Strategic Human Resources Planning and Management" (e.g. Rothwell and Kazanas, 1988). Tsui (1984, 1987) has described the range of activities involved in human resource management and the differing expectations of the department's constituents. For the most part, however, literature on "strategic human resource management" (SHRM) is prescriptive, vaunting the numerous advantages possessed by companies which take this approach to P/HR. Especially lacking is a theoretical framework to describe this transition.

This paper suggests that the present and past development and growth in personnel/human resource management has been due to a recognition by organizations that

an increasing complexity and concomitant uncertainty in the human resource arena needed to be addressed (e.g. Dyer, 1985; Ledvinka, 1982; Ritzer and Trice, 1969). This conceptualization suggests that the P/HR function can be considered a boundary spanning unit, charged by the organization with the role of addressing human resource uncertainty (Thompson, 1967). Granted, one criticism of the research reviewed above was its choice of a sample that consisted of people that were "by definition" boundary spanners. The review of P/HR literature that follows is included to support, on a theoretical level, the use of P/HR departments as boundary spanning units. The above-mentioned research did not discuss their choice of sample in terms of Thompson's theory or any other. Second, as will be discussed in the Methods section, the analysis allows boundary spanning to vary, as opposed to considering it as a given. These two components of the present test of Thompson's theory should, in combination, address the weaknesses identified in the previous review.

The recent transition in P/HR is only the latest in the development of P/HR management. P/HR has an interesting history, much of which has been described by Ritzer and Trice (1969), Jacoby (1985), and Baron and his colleagues (Baron, Dobbin, and Jennings, 1986; Baron, Jennings, and Dobbin, 1988). Ritzer and Trice (1969) and Jacoby (1985) argue that "welfarism" and "scientific management" were two factors that contributed to the development of personnel management in the years prior to World War I. The latter, labelled "Taylorism", focused on the importance to organizations of attracting and selecting the "best" employees and placing them in the job for which they were best suited. The influence of the former is seen as due to a general trend towards paternalism in society and the passage on the state level of workmen's compensation laws. At this early stage in the development of the field, considerable attention is given to forces in an organization's environment for the establishment of personnel functions.

Between World War I and World War II, Ritzer and Trice (1969) identify four environmental forces that further reinforced the development of the personnel function. First, the administration of psychological tests developed by industrial psychologists to further aid organizations in selection of employees often fell to newly established personnel departments. Second, the 1930's were a decade of increased unionism, marked by the passage of the Wagner Act and the founding of the CIO. Many organizations, especially larger ones, developed departments to address these areas, usually within personnel. Third, in response to the increasing influence of unions, organizations began to develop considerable benefits packages, dubbed the "American Plan". These, too, were generally administered by an existing or newly created personnel function. Finally, increasing government regulations, such as the Social Security Act of 1936 and the Fair Labor Standards Act of 1938 forced management to develop justifiable compensation programs. Again, this responsibility was generally given to the personnel function, thereby further expanding and specializing the department's complexity. In terms of Thompson's theory, these forces described by Ritzer and Trice (1969) are representative of the kinds of events that create uncertainty for organizations, leading in theory to the development of boundary spanning units, represented here by the introduction and development of personnel administration.

In their examination of the development of personnel practices between 1927 and 1946, Baron et al (1986) focus on three constituencies, all external to the organization: labor unions, personnel professionals, and the state. The practices which were considered specifically are the specialization and routinization of work roles based on the doctrine of scientific management and the development of personnel practices designed to promote long term employment. Edwards (1979) would consider the former as a sign of technical control and the later one of bureaucratic control. In examination of archival data, Baron et

al (1986) note a rapid increase in bureaucratization during World War II in both large and small companies. While size was strongly correlated with the presence of bureaucratic procedures, as Blau and Schoenherr (1971) would predict, there were many exceptions. Baron et al (1986) found that as time passed, size became much less useful as a predictor of the presence or absence of personnel procedures, corroborating the institutionalization hypotheses of Tolbert and Zucker (1983). In essence, the institutional argument proposes that with time, practices become "widely understood to be a necessary component of rationalized organizational structure" and therefore more widely adopted, regardless of organizational size (Tolbert and Zucker, 1983: 35). The second variable which had been hypothesized as a reason for the development of personnel practices was turnover. Baron et al (1986) found little support for this hypothesis. In their sample, a consistency in procedures across industries existed, while rates of turnover varied greatly.

Baron et al (1986) explained the bureaucratization in personnel practices through a description of the role of the federal government, a major external force, during and after World War II. First, War Labor Board rules regarding employment stabilization plans required a structure to personnel practices that encouraged isomorphism between employers. Secondly, government forces on employers to justify manpower encouraged organizations to cut turnover. Baron et al (1986) suggest that while these pressures were present for all employers, the increased visibility of large employers to regulatory agencies made the further bureaucratization of their personnel function more likely. When the war ended and the constraints imposed by the wartime commissions were removed, personnel specialists needed to justify their continued existence. In order to build on their past experiences in a way that would seem valuable to organizations, personnel specialists emphasized first their abilities to attract and accurately evaluate employees, and then their abilities to enhance productivity and manage employee relations (Hayes, 1945; Kirkpatrick,

1945). Not only did the personnel field continue to experience rapid growth after the war, but the bureaucratic personnel practices developed during the war continued to survive and diffuse, further promulgating the existence of the personnel specialty (Baron, Jennings, and Dobbin, 1988).

While the transition that Baron and his colleagues describe focused on the establishment and survival of P/HR, the essence of the transition P/HR is experiencing in the 1980s centers on the proactive role that P/HR departments should be shaping. The assumption is that through the enactment of a future oriented P/HR department, an organization will be more competitive in business (Misa and Stein, 1983; Burack, 1986; Flamholtz, Randle, and Sackmann, 1987a, b). The reasons given for the relatively recent recognition of the effect P/HR can have on corporate success are of several types. For example, Meyer (1976) and Layton (1987) give credit largely to the P/HR field itself for enlightening management to the abilities of its P/HR department. The other common explanation is that management realized that in order to successfully respond to continual change in the social, legal, and economic environments, a more active P/HR function would be necessary (Devanna, Fombrun, and Tichy, 1981; Carroll and Schuler, 1983; Burack, 1986). A recent survey of top managers by the American Society of Personnel Administrators (ASPA) found that the current importance given to P/HR was explained by the role the department could have in addressing important issues such as high labor costs, low productivity, quality of worklife issues and compliance with government regulations (Foltz, Rosenberg, and Foehrenbach, 1982).

In their review of these recent changes in P/HR, Rothwell and Kazanas (1988) cite two sources for the current focus on P/HR. One source comes from the P/HR field itself, striving to enhance their status as a profession (Schlesinger, 1983). The second source is top management, which has somehow come to realize that P/HR can make a substantive

contribution to the operation of the business (Foltz et al, 1982). In setting the context for the transition in P/HR, Rothwell and Kazanas (1988) note that the employment environment has become much more complex. No longer can employers assume that appropriately skilled labor is readily available. Rothwell and Kazanas (1988) cite three indicators of the increasing complexity of the employment environment. First, there are fewer people working in unskilled jobs. Second, human factors such as creativity have become more important in remaining competitive. Third, the U.S. economy has become more service oriented, meaning organizations are more labor-intensive. In addition, work itself has become more complex, further driving the need for a coordinated human resource effort. For example, technology has created, as well as solved, problems. The implementation of technology creates the P/HR problem of skill obsolescence among the current workforce.

Another source of complexity for the P/HR field comes from the increasing government regulations regarding personnel practices. Rothwell and Kazanas (1988) identify three major areas of governmental influence: (1) Equal employment opportunity (EEO) and affirmative action, (2) Labor laws (such as the Taft-Hartley Act of 1947 and the Labor Management Relations Act of 1974), and (3) increasing restrictions in the interpretation of the "employment-at-will" principle. A final source of complexity for the P/HR field noted by Rothwell and Kazanas (1988) concerns the changing demographics of the U.S. population. The most significant of these is the aging of the workforce. One special problem this will create is a "log-jam" of sorts at the middle management level. A future P/HR challenge will be to successfully manage and motivate employees who have little hope for promotion, due to opportunities blocked by the large numbers of older, established workers. While the above documents change in the P/HR environment over the past few decades, there have been some important changes that have occurred in just the

past few years. Most significant has been the role of the Reagan administration that has, through decreased government involvement, reversed a trend that had lasted nearly 50 years (Odiome, 1983). The clearest example of these policies is the decrease in influence and power held by the Occupational Safety and Health Administration (OSHA) as measured by cuts in budget and staff (Odiome, 1983).

No one appears to be arguing against an expanded or redirected P/HR role. At the same time, it is clear that that role has not come to be adopted universally by organizations (Devanna *et al* 1981; Vicere, 1987). The questions of interest then become under what conditions do P/HR roles develop and under what conditions are such roles beneficial to the organization? The former question is addressed to some extent by Devanna *et al* (1981). They argue that many companies have not approached P/HR in a strategic way because: (1) management does not feel such a role is necessary; and (2) the P/HR department does not possess the skills to carry out a strategic role. On the other hand, Misa and Stein (1983) found that productivity leaders were also leaders in P/HR, with P/HR executives in leading companies five times more likely to state they were included in organizational decision making. P/HR executives in the "non-leading" companies reported that their role was concentrated on hiring, firing, and policy administration. It is important to note that a question of causality concerning the above relationship was not addressed in the study.

Mahoney and Deckop argue that the changes in the way P/HR is perceived from personnel to human resources, administration to management, and from human relations to organizational effectiveness have occurred eclectically, without the benefit of an overriding conceptual framework. While work in scientific management (e.g. Taylor, 1911), industrial psychology and industrial sociology (e.g. Roethlisberger and Dickson, 1939) clearly had roles in the early development of the field, none provided an overall model.

Instead, pieces of each were applied to personnel issues, such as work on job design (scientific management), learning curves (I/O psychology), and social interaction (industrial sociology). Since the years prior to World War II, P/HR has continued to develop without a conceptual base and continues to consist of a collection of activities associated with a model or theory from another field (Mahoney and Deckop, 1986).¹

Summary. From this review of the literature describing the transition from personnel administration to strategic human resource management, two themes emerge. The first theme, present in nearly every presentation of the subject, is that contingencies in the environment have been the major force driving the establishment of and change in P/HR. The second theme, which underlies most work and is explicitly recognized in other work, is that a unifying theoretical framework to explain the transition is missing. In this paper, it is argued that Thompson's open systems concept of a boundary spanning unit provides that framework. The work by Thompson (1967) and others provides a solid theoretical basis from which to generate hypotheses concerning change in P/HR management.

The P/HR Department as Boundary Spanner

While the term "boundary spanner" itself has not been used to describe P/HR departments, the concept represents what has been discussed by various authors under various labels. Carroll and Schuler (1983), Dyer (1985), and Rothwell and Kazanas (1988) discuss the role of the "environmental scanner" as one of many in a system of roles carried out by the P/HR function. Milkovich, Dyer, and Mahoney (1983) note that P/HR

¹ There has been some study of the development of personnel as an occupation along the lines suggested by Hughes (1958) and Wilensky (1964)(e.g. Ritzer and Trice, 1969; Watson, 1977). Generally, the questions addressed in this work concern the process by which individuals stake claim to an area of expertise and professionalize.

departments carry out an "external analysis", which involves the analysis of factors like government regulations, social values, and labor market conditions. Milkovich et al (1983) describe a four step process in external analysis that is similar to the boundary spanning process described above. In their model, the P/HR function monitors the environment, screens the information, communicates the results internally, and develops plans for action. In their development of the human resource management system, Scarpello and Ledvinka (1988) give significant attention to the role of the P/HR function in helping the organization address issues in the employment and regulatory environments. In this conceptualization, the P/HR "subsystem" is one of several subsystems which interact in an effort to protect the organization's production process. Heneman, Schwab, Fossum, and Dyer (1986) also present a model of P/HR that incorporates environmental components including economic, labor market, and regulatory forces. As noted by Milkovich et al (1983), the external scanner role played by P/HR departments has been theoretically well elaborated, but has been the subject of little empirical work. This article reports on a study that uses the boundary spanning framework to test hypotheses regarding the role of P/HR in addressing issues in the external environment, and discusses those findings in regard to explaining the transition that has occurred in P/HR.

This analysis argues that the environment in which P/HR boundary spanning occurs can be divided into three sectors. The task environment sector is made up of the traditional responsibilities of a P/HR unit, such as staffing the organization and developing and administering compensation policies. In Thompson's (1967) words, these functions represent the core technology of a P/HR department. External functions in this environment include, for example, analysis of labor market conditions (Scarpello and Ledvinka, 1988; Milkovich et al, 1983). Since these functions provided the impetus for the development of the P/HR field and have been the core technology of P/HR departments for

years (Baron et al, 1986), Thompson's theory and institutionalization theorists would suggest that these functions have become routinized and procedures developed to address most of the contingencies that would arise.

A second environmental sector in which P/HR departments boundary span is the regulatory environment (Heneman et al, 1986; Scarpello and Ledvinka, 1988). In this sector, P/HR units monitor legislative changes and interpret them for the organization. Internally, P/HR departments work to insure organizational compliance with regulatory forces, such as the Occupational Safety and Health Act (OSHA) and Equal Employment Opportunity (EEO). While one might expect that regulations serve an uncertainty reducing function, thereby lessening the need for boundary spanning to occur, they are in fact often ambiguously worded and subject to multiple interpretations by individuals, organizations, and courts (Scott, 1987). In addition, new regulations are constantly put in place, increasing the importance of boundary spanning in the regulatory environment. Three examples of recent legislative actions are: (1) the Immigration Reform and Control Act of 1986 which makes the employer culpable for employing illegal aliens (Berry and Appleman, 1987; May, 1987; Sullivan, 1987); (2) the Consolidated Omnibus Budget Reconciliation Act of 1985 (COBRA) which requires employers to, among other things, provide certain former employees and their families with health coverage (Murphy, Barlow, and Hatch, 1987); and (3) the Plant Closing Act, which would require employers to give a 60 day notice to employees before closing a worksite. The continual change in and the sometimes unclear implications of regulation, in conjunction with the heavy costs a violating organization could face, make the monitoring of this environment important. While this is also a function that has been performed by P/HR departments for some time, it is an area in which the organization has less direct influence and which can have costly

implications for industry (Ledvinka, 1982). As a result, one would expect high degrees of uncertainty to be present.

The third environmental sector in which P/HR departments span boundaries is what this paper calls the P/HR social environment (e.g. Milkovich *et al*, 1983). The social environment is made up of the complex web of societal issues which may affect worksites. At present, visible issues in the social environment include: alcohol and drug abuse (Blum and Roman, 1988), drug testing/screening (Blum, 1988), stress (Markowich, 1987), health promotion and wellness (Aberth, 1986; Sherwood, 1986), and AIDS (Levine, 1986; Letchinger, 1986; Myers and Myers, 1987; Waldo, 1987). These issues create P/HR uncertainty which can disrupt the core technology of the P/HR department by, for example, inhibiting the attainment and maintenance of an appropriately qualified workforce. This disruption of the P/HR core has the potential to create disturbances of the organization's core technology. As noted earlier, the monitoring of this environmental sector has also been a long standing concern of P/HR departments. However, it is suggested that this sector is also the most rapidly changing. For example, AIDS in the workplace is an emerging issue with characteristics that make it unlike any other the field, or the workplace, has faced. As Duncan (1972) argued, organizations have a more difficult time reacting to change than to complexity.

Each of these three sectors contain uncertainty which has varying degrees of relevance to an organization. To the degree that uncertainty is present, Thompson's (1967) theory would suggest boundary spanning will occur. Daft *et al* (1988) have argued that in addition to the presence of uncertainty, top management must feel the issue or sector has significance for the organization. As a result, they consider not only Duncan's (1972) formulation of uncertainty (complexity and rate of change), but also the importance of a sector to an organization in their concept of "strategic uncertainty". Boundary spanning can

be driven by any one or a combination of the complexity, rate of change, or importance of an environmental sector. For example, in the social sector, the organization is confronted with complex and often rapidly changing issues, but top management may not feel these issues are of concern to the organization. Organizations in isolated, rural areas may feel AIDS is not a threat to the workplace (low importance), but agree that it is a complicated and changing issue.

A second factor that influences boundary spanning is the degree to which an organization has an agreed upon method for addressing the environmental sector in question. In areas where a high degree of consensus concerning the "solution" the organization should adopt to address the environment, the organization will be less dependent on boundary spanning. Where no such consensus exists, boundary spanning is needed. In Thompson's (1967) words, the development of a strategy to address uncertainty represents a modification creating structure that matches that of the environment, thereby reducing the dependence of the organization on boundary spanning activities. The incorporation of these technologies to address uncertainty into the everyday operation of the organization is a central tenet of institutionalization theories (e.g. Tolbert and Zucker, 1983).

Two final points concerning the relationships between environmental sectors are important to note. First, components of the task environment are explicitly influenced by constraints in the regulatory environment. For example, a workforce reduction (a task environment issue) is most feasibly and defensibly accomplished when aspects of the regulatory environment (EEO and Affirmative Action requirements) are kept in mind. Clearly, the elements of the social environment can have bearing on elements of the regulatory environment, too. For example, employers are now becoming concerned with the possibility of stress related disability claims (Voluck and Abramson, 1987), and the

legal ramifications of the way employees with AIDS, as well as their co-workers, are treated (Wing, 1986). Second, history indicates that issues can evolve, progressing from one sector to another. Typically, social issues (e.g. discrimination) become regulated (Affirmative Action) and can become a part of the P/HR department's core tasks.

General Model and Hypotheses

This dissertation examines the way people in P/HR departments perceive their environments in terms of uncertainty and how they respond to reduce that uncertainty through interaction with organizational and environmental constituents. The model and method parallels and expands on that in the study of Chief Executive Officer environmental scanning by Daft *et al* (1988) which was summarized earlier. The starting point is similar to that tested by Daft *et al* (1988). Hypothesis One is concerned with the levels of uncertainty perceived in sectors of the P/HR environment. In the task environment, the department is engaged in its most basic, routinized work. Therefore, uncertainty is expected to be low. In the regulatory sector, the costs of misreading the environment are greatest. As a result, perceived uncertainty in this sector is expected to be greatest. The social sector is less directly related to the organization's core, but is suggested to be the least familiar and least controllable to organizational members. Uncertainty in this sector is expected to fall, on average, between the task and regulatory sectors.

Hypothesis One: Environmental sectors will differ in the amount of perceived uncertainty reported by the P/HR department, with those sectors having the greatest potential consequence for the organization's technical core containing the highest uncertainty. Specifically, the task environment will display the least uncertainty, followed by the social environment and the regulatory environment.

The box in Figure 1 describes the model of boundary spanning introduced by Thompson (1967) and elaborated by others (e.g. Tushman and Scanlan, 1981a, b). In the first portion of the box, the external linkages maintained by boundary spanners are hypothesized to be reciprocally related to top management's perceptions of environmental uncertainty. This reciprocal relationship is complicated, and as such needs further elaboration. According to Thompson's (1967) theory, organizations establish boundary spanners in response to perceptions of uncertainty. These boundary spanners then monitor the environment for the organization. The result of this activity is either a reduction in uncertainty as the boundary spanner learns about the environment or an increase in uncertainty when the boundary spanner finds out things are much worse than anticipated. The former scenario would suggest a (+) (-) pattern be assigned to the paths in Figure 1, while the latter would suggest a (+) (+) is appropriate. To further complicate matters, in the latter scenario Thompson would suggest that over time, the boundary spanner would achieve some mastery over the environment allowing the organization to develop institutionalized mechanisms to reduce or cope with uncertainty. For example, an organization that vertically integrates a producer of its raw materials, thereby bringing it into the organization, reduces its uncertainty. This would clearly result in a different pattern of boundary spanning than would be expected in an organization that adopted a coping

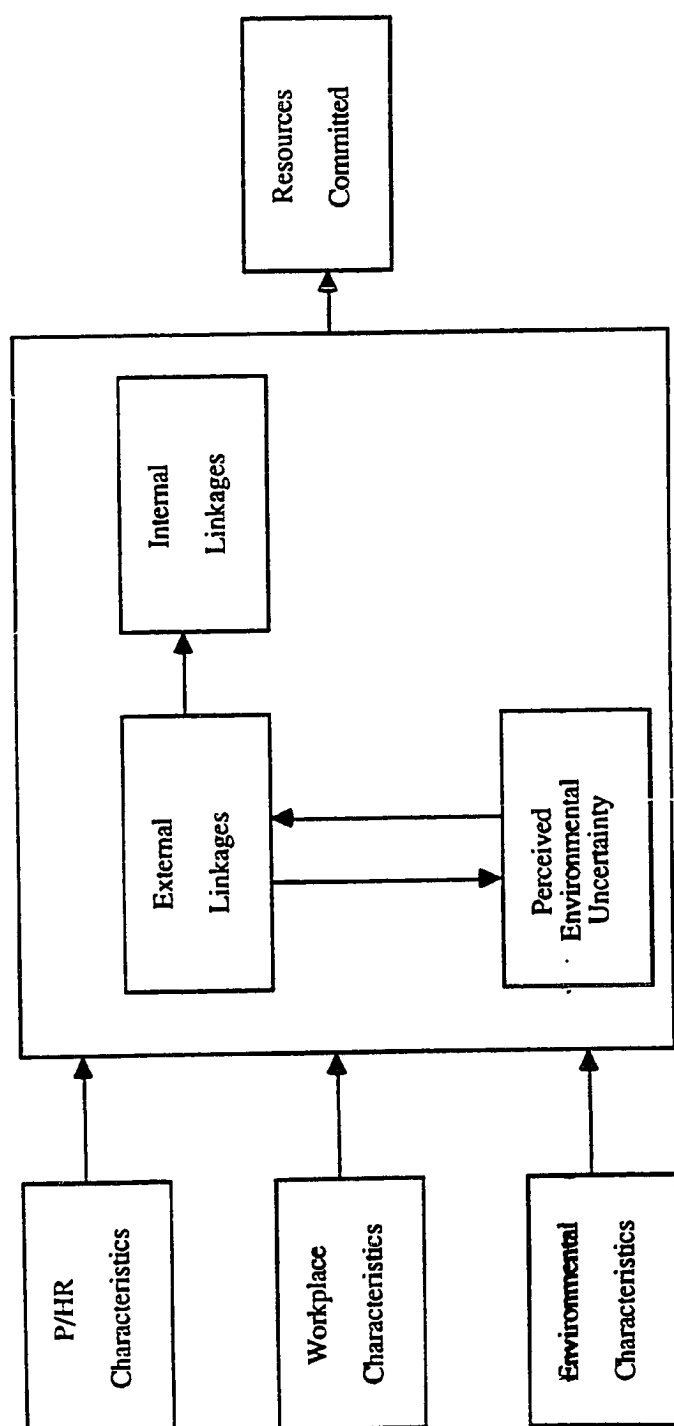


FIGURE 1
General Model Illustrating the Boundary Spanning Process
and Suggested Antecedent Variables

strategy for addressing uncertainty, such as stockpiling of resources. In the former case, boundary spanning may cease, while in the latter it would continue, despite the fact that in each case, uncertainty had been reduced. While these relationships are theoretically defensible, they are extremely difficult to operationalize. This research has tried to do so by incorporating sectors that at one time led to the establishment of the personnel function which may have stabilized (e.g. regulatory and task sectors) and new areas which, it is suggested, have not (e.g. social sector).

Hypothesis Two suggests that in sectors with high amounts of perceived uncertainty, the respondent will report a high degree of boundary spanning activity. In turn, those departments displaying high boundary spanning activity are expected to perceive greater uncertainties. This hypothesis is also influenced by the work of institutionalization theory. The boundary spanning - perceived strategic uncertainty link is also influenced by the degree to which a consensus exists among management as to how to best address the environmental sector in question. Where no consensus exists because a situation is complicated, rapidly changing, or novel, the boundary spanning - perceived uncertainty link is expected to be stronger, as organizational members search for a solution. In cases where uncertainty is high but consensus on the technologies to employ exists, than a weaker boundary spanning - perceived uncertainty link is expected. In the social sector, where technologies to address uncertainty are suggested to be missing or immature, this relationship is expected to be strongest. In the task sector, where practices have been institutionalized, the relationship is expected to be weakest. In the regulatory sector, practices have been institutionalized to a lesser degree because of the continual change in the sector. As a result, the boundary spanning - perceived uncertainty link is expected to be slightly stronger than that in the task sector. This leads to the second hypothesis tested in the dissertation:

Hypothesis Two: Boundary spanning (external linkages) will be positively and reciprocally related to perceptions of uncertainty in the P/HR environment. The strength of this relationship is related to the sector of the environment being considered, with the strongest relationship present in the sector with the least developed technologies (social), and the weakest relationship in the sector with the most highly developed technologies (task).

The next relationship to be discussed is based on the assertion by Tushman and Scanlan (1981a) that boundary spanners are well connected, both inside and outside the organization. One of the criticisms of earlier boundary spanning research made in this dissertation concerned the lack of attention paid to what happened to the information obtained by the boundary spanner. The definition of boundary spanning developed above stressed that at least some of the information collected was of some consequence for the organization. Clearly, since boundary spanners play a filtering role, not all information is to be passed on, but some must. This dissertation examines the relationship between reports of boundary spanning and the degree to which the P/HR department is involved with other organizational units in addressing the information obtained through boundary spanning strategies. This is the second relationship portrayed in the box at the center of Figure 1. again, in sectors where the organization does not have technologies to address uncertainty, a stronger external linkage - internal linkage relationship is expected, as the dissemination of information gained through external linkages becomes more valuable. In routinized areas, internal linkages are expected to be less important. Hence, the third hypothesis tested in the dissertation is:

Hypothesis Three: P/HR departments that engage in high levels of boundary spanning activity will also display high levels of interaction with other organizational units concerning these issues, supporting the existence of strong internal, as well as external, linkages as suggested by Thompson (1967) and Tushman and Scanlan (1981a). Specifically, the correlation between external linkages and internal linkages will be strongest in sectors with the least developed technologies (social) and weakest in sectors with the most highly developed technologies (task).

The existence of internal linkages does not in itself reveal anything about the organization's reaction to the boundary spanner. It is possible for a boundary spanner to be highly active through both external and internal linkages, but to be providing the organization with information of minimal value. On the other hand, an efficient boundary spanner might be able to accurately evaluate the environment with a minimum of scanning activity. Ideally, an analysis testing these relationships might include data from those in the organization to whom the boundary spanner reports. For this analysis, a proxy measure of organizational resources committed to address P/HR uncertainty is used to examine the relationship between boundary spanning and organizational efforts to create consensus strategies with which to address uncertainty. This relationship is outlined in Hypothesis Four.

Hypothesis Four: Organizational resources to address uncertainty in the P/HR environment will be explained by the maintenance of external linkages by the boundary spanner.

One last concern which needs to be addressed concerns the three categories of independent variables on the left portion of Figure 1. These remaining variables are used to develop a model which can partial out the reciprocal relationship between perceived uncertainty and external linkages discussed above and were selected based largely on their importance to other models in organizational theory. First, workplace characteristics, such as size and number of sites visited are considered. These are all variables that were controlled for in the Daft *et al* (1988) study and have been shown by a number of researchers to have broad, overriding effects on the relationships between other organizational characteristics (e.g. Blau and Schoenherr, 1971). It is also argued that characteristics of the P/HR department have an influence on the amount of boundary spanning which occurs. For example, P/HR departments that have taken on the proactive role described above are expected to engage in greater amounts of boundary spanning. Finally, environmental characteristics that affect the organization at large, such as concerns about competition, are suggested to have an effect on the context in which the P/HR department operates, and as such on the amount of boundary spanning and perceived uncertainty that occurs in the P/HR function.

CHAPTER III

METHODS

Sample

The data for this analysis were collected from a sample of Atlanta area worksites with at least 250 employees. Lists of firms for the 18 county Atlanta Standard Metropolitan Statistical Area (SMSA) were compiled from Duns Marketing Identifiers, the Georgia Manufacturing Directory, and Chamber of Commerce lists from each county. These lists were cross checked in the development of the population frame. From the population of 649 companies, a sample of 350 was drawn at random. Once selected, companies were contacted to verify the number of employees. Companies which turned out to have less than 250 employees or could not otherwise be accounted for were replaced by a company similar in size and type, with type measured by industrial classification. Once the sample had been selected and confirmed, companies were contacted and interviews scheduled with the highest level human resource person at that site. Within the sample of 350, companies were selected at random for inclusion. Of the 170 companies contacted for interviews 142 (83.5%) agreed to participate. Comparisons of the participating companies with those who declined to participate did not reveal any systematic differences in terms of company size or industry sector, two variables easily obtained for non-participants.

Instrumentation

Interviews

The interviews were semi-structured based on a pre-tested interview schedule and lasted an average of approximately 90 minutes. The interview was developed based on a review of the P/HR and boundary spanning literatures and was reviewed by an expert panel of academicians and revised based on their feedback before use. After this revision, the interview was pretested on a random sample of companies which did not fall into the study sample. This process led to further revisions based on the quality of information obtained in the interview and the pretest respondent's comments concerning its structure and language.

Mailback Questionnaire

In addition to the interview schedule, data were collected by the means of a mailback questionnaire left with respondents at the close of the interview. The mailback questionnaire consisted of largely original measures based on the P/HR literature and was itself reviewed and critiqued by a panel of academic experts. Based on their comments, revisions were made leading to the final version of the questionnaire. After follow up reminders through the mail, 124 (87.3%) of the interviewed companies returned completed questionnaires. Respondents and non-respondents to the mailback questionnaire were compared using data collected in the interview. When considering variables such as industry type, number of employees and size of P/HR department, no significant differences were found.

Issues of Coder Reliability

Concern about the reliability of measurement is a problem that can arise when multiple interviewers and interview coders are used. Several steps were taken in the data collection and data reduction processes in an effort to ensure the reliability of the data.

While neither of these processes provides a guarantee of high reliability in the interview data, both eliminate a considerable amount of variation due to differences in either the interviewer or coder. First, all interviewers were familiarized with each of the topics covered in the interview instrument. Second, all interviewers observed approximately six interviews conducted by the author. After this period of observation, interviewers conducted two or three interviews under the observation of the author. In both of these phases, the author and the interviewer being trained met after each interview to compare notes and discuss the interview in general. The main focus of these discussions was to sensitize the interviewers to the significance of the responses.

In the data reduction process, responses to the open ended interview questions were coded into a series of multiple choice variables designed to capture the range of responses given by the P/HR managers but also to provide a categorization scheme. Where possible, the person responsible for conducting the interview also did the coding. In some cases, other individuals were trained to code the interviews. The training for this task consisted of coding an interview which was also coded by the interviewer. Discrepancies in coding were discussed and resolved. Throughout the coding process, coders were responsible for doing blind reviews of each other's coding and resolving any conflicting codes with the help of the interviewer, if necessary.

Measures

Before discussing the measures in detail, it is important to take into consideration an important debate in the literature concerning measurement of the "environment". For good reason, there has been some debate concerning whether the use of objective or perceptual measures of the environment is more appropriate. Most have argued that perceptions are most important, since an individual must perceive some phenomenon in order to consider it (Dill, 1958; Lawrence and Lorsch, 1967; Duncan, 1972). On the other hand, Pfeffer and

Salancik (1978) argue that objective characteristics of the environment obviously have an impact on the success or failure of organizational actions, and need therefore to be considered. Scott (1987) reviews this literature and suggests that when models are concerned with choices of organizational members then perceptual measures of the environment are appropriate. When models are concerned with the outcomes of choices, then objective measures are appropriate because outcomes are effected by a number of forces beyond the perceptions of organizational members (Scott, 1987: 136). Following Weick (1979) and Scott (1987), this paper focuses on the enacted environment: that created by organizational members based on the 'objective' characteristics of their surroundings. The consequence of this orientation for the analysis is that measures of perceived environmental characteristics.

Dependent Variables

Perceived Environmental Uncertainty: Measures of environmental uncertainty in each of the three environmental sectors were developed based on work by Daft et al (1988). In their analysis of CEO environmental scanning, they develop measures of strategic uncertainty based on Duncan (1972). While Duncan (1972) argued that uncertainty was made up of the complexity and rate of change in an area (i.e. $U = C + R$), Daft et al (1988) suggest that the strategic uncertainty present in an environmental sector can be estimated as the product of the importance of a sector with the sum of its complexity and rate of change ($SU = I(C + R)$). The importance component is included as a predictor of the "need for information impinging upon" organizational members (1988: 130). In this analysis, perceptions of importance, complexity, and rate of change are measured for P/HR areas hypothesized to represent each of the three environmental sectors identified above: task environment, social environment, and regulatory environment. Task environment measures include staffing, developing and administering the performance appraisal system,

and developing and maintaining compensation policies. Regulatory environment measures include employee safety, assuring compliance with government regulations, employee relations, and health care cost containment. Social environment measures include dealing with stress in the workplace, addressing issues about AIDS in the workplace, responding to the impact of alcohol abuse on the workplace, responding to the impact of drug abuse on the workplace, responding to the impact of family problems on the workplace, and promoting healthy lifestyles. For each of these measures, the respondents were asked to indicate on a 1 to 5 scale the importance, complexity, and rate of change present. The analysis will examine Duncan's (1972) conceptualization of uncertainty ($C + R$) and then through the consideration of an interaction term in a multiple regression ($(I)(C+R)$) the "fit" of the Daft *et al.* conceptualization of strategic uncertainty. Finally, the additive model ($I + C + R$) is also tested.

Boundary Spanning - External Linkages: For each of the areas discussed above, the respondents were asked to indicate in the mailback questionnaire (also on a 1 to 5 scale) the extent to which they addressed each of the issues noted above by working with experts outside the organization.

Boundary Spanning - Internal Linkages: Several measures of this second stage of the boundary spanning process were obtained through both the mailback questionnaire and the interview. On the mailback questionnaire, respondents were asked to indicate (on a 1 to 5 scale) how often they worked with other departments inside the organization in addressing issues in each of the sectors.

Resources Committed: In the interview, respondents were also asked to report on the extent to which their organization had committed resources to address issues that affected their employees. The issues focused mainly on aspects of the social sector, and are listed in Table 1. It is suggested that the decision to commit these resources is based on

TABLE 1
Scales Used in the Analysis:
Items and Measures of Internal Consistency

The first four of the scales are based on responses to the following instructions: "To what extent do you think top management sees the following as problems for your organization?" (Scaled 1 to 5)

EXTENV (alpha = .65) These items are concerned events in the external environment.

1. Rapidly changing social climate
2. Rapidly changing political climate
3. Impending legislation

INTENV (alpha = .79) These items are concerned with disruptions in the employment environment.

1. Absenteeism
2. Worker's compensation claims
3. Accident rate

CORENV (alpha = .61) These items are concerned with threats to the organization's core technology.

1. Rapidly changing technology
2. The competition gaining ground
3. Unstable demand for your products

PHRENV (alpha = .78)

1. Available supply of adequately trained employees
2. Turnover in the workforce

TOPMGT (alpha = .77) This scale is the sum of responses to the prompt: "Based on your expertise, indicate whether your department exceeds, meets, or fails to meet top management's expectations in each of the following areas" (Scaled 1 to 3)

1. degree of innovation in P/HR management
2. skills and expertise held by P/HR department members
3. effectiveness in dealing with personnel regulations
4. role of P/HR department in organizational decision making

RESOURC (alpha = .71) measures the degree to which the organization has committed tangible resources in areas involving the four sectors. It is computed as the sum of the following items, where a 1 is coded if resources have been committed and a 0 if they have not, as reported in the open ended interview.

1. Has the organization committed resources for dealing with alcohol problems in the workplace?
2. Has the organization committed resources for dealing with drug problems in the workplace?
3. Does the company do pre-employment drug screening of job applicants?
4. Has the company committed resources for dealing with employee stress?
5. Has the organization committed resources for dealing with day care issues in the workplace?
6. Has the organization committed resources to address AIDS in the workplace?
7. Does the company have an EAP?

previous boundary spanning activity by the P/HR department. Since they focus on social and task sector issues, they should be related to reports of boundary spanning in the social sector and unrelated to boundary spanning in the regulatory sectors.

Independent Variables - Perceived Environmental Characteristics

Four of the independent variables were taken from a question on the mailback questionnaire which measured the extent to which the respondent felt top management was concerned about a number of issues (on a scale of 1 to 5 where 5 represents a high degree of concern). The description of these measures follows, and is repeated in Table 1 with mnemonics used in presenting the results.

External Social Climate (EXTENV) was measured using three items ($\alpha = .65$) addressing concern about the rapidly changing social climate, the rapidly changing political climate, and impending legislation facing the firm.

Personnel/Human Resource Management (INTENV) concerns about the internal environment were measured with three items ($\alpha = .79$) assessing top management's concern with the organization's rate of absenteeism, worker's compensation claims, and accident rate.

Core Technology (CORENV) concerns were also measured with three items ($\alpha = .61$) which focused on top management's concern about the impact of external events on the organization's core technology: rapidly changing technologies, competition gaining ground, and unstable demand for the organization's product.

Independent Variables - P/HR Characteristics

Top Management Evaluation of P/HR (TOPMGT) was measured using four items ($\alpha = .77$) which asked the respondent to report (on a 1 to 3 scale) his/her perceptions of management's opinion of the professionalism of the P/HR function. Items included: degree of innovation in P/HR management, the skills and expertise held by P/HR

department members, the department's effectiveness in dealing with personnel regulations, and the role of the P/HR department in organizational decision making.

In addition, P/HR characteristics measured in the interview were included in the analysis. P/HR Role is a variable created to describe the role played by the P/HR department in the organization, as reported by the respondent. The measure ranges from 1 to 5, with a 1 indicating the department plays a strictly administrative role and a 5 indicating a strictly consultative role. Departments that play an administrative role are not expected to engage in much boundary spanning, as their focus is on enforcement and interpretation of existing policy. A consultative department, however, would be expected to boundary span, since their role involves the provision of advice on P/HR issues. These two extremes can be illustrated by considering an organization where the corporate headquarters engages in considerable boundary spanning and develops policies and procedures based on the environment which a branch site, playing an administrative role, merely interprets. P/HR Emphasis is a variable that measures the interviewer's impression of the degree to which the department focuses on traditional issues and methods versus "new" issues and innovative methods. For example, a P/HR department is generally concerned with selecting the best applicants possible but can do that based on intuition or through the use of a battery of psychometric tests. The implication here is that a traditional focus implies that an organization is staying with what is familiar. This does not require boundary spanning. An organization that uses innovative methods cannot find out about their existence without boundary spanning. Organizations which rely on the former would be classified as traditional and those who rely on the latter would be classified as progressive. It is important to note that no effort or value judgement is made to determine whether one or the other approach is more appropriate or successful. Instead, the emphasis is on creating a classification based on the literature concerning progressive P/HR departments. P/HR

Stance is also based on the interviewer's impressions of the P/HR department. The variable is a dichotomous one, with a code of 1 indicating a reactive stance and a 2 indicating a proactive stance.

Independent Variables - Workplace Characteristics

Finally, five objective measures of workplace characteristics were collected and used as independent variables in the analysis. Respondents were asked to report the number of sites that the company operated and the number of employees employed by the company overall. Three variables were used to assess the current situation regarding employment trends at that location specifically. Respondents were asked to describe the situation at their site regarding growth or loss of employees and profits (1 = shrinkage, 2 = stable, 3 = growth) and the general trend in employment and profits (1 = steady growth trend, 2 = stable, 3 = steady decline). Turnover was measured as the percent of the workforce that was replaced in the past 12 months. Because the turnover variable was positively skewed, it was transformed by the natural log before inclusion in the analysis. Union presence or absence is a dummy variable where a "1" was coded in unionized companies and a "0" otherwise. The P/HR department's relationship to a corporate program is coded as an ordinal variable where a "1" is used to describe a site that is entirely dependent on corporate headquarters for policy and procedures, a "3" indicates that a site is entirely autonomous or has no relationship to a corporate headquarters (i.e. one site only).

Analyses

There were three major components to the analysis. First, issues of response set bias were considered. Second, the three sector conceptualization of the P/HR environment was examined empirically. Each of these analyses were considered before moving on to the third portion of the analysis which involved the test of Hypotheses One through Four.

Measurement Issues

A concern of any analysis, especially one that relies heavily on a closed ended questionnaire as a source of data, is response set bias. Response set bias is a common problem and is said to exist when responses are attributable more to the format in which answers were requested than to the stimuli with which the respondents were presented (Bailey, 1978). In closed ended questionnaires, response set is always likely to be present so the issue becomes the degree to which a bias exists. The first part of the analysis is concerned with issues of response set bias and tries to assess the degree to which it might contaminate the analysis.

While it is at best difficult to know when a problem with response set bias exists, there are a few steps which can be taken to examine the degree to which the data may be affected. Responses to the questionnaire items were examined, both across and within-subject, in an effort to identify problems. It is important to note that where possible, data collected through other methods (interview) and objective data (e.g. turnover, number of employees) has been included in the analysis, and while still based on self report, is an effort to reduce the effects of response set bias. Given the format of the mailback questionnaire which provides a considerable portion of the data in this analysis, there are two ways in which response set could be a problem. The mailback instrument included thirteen sections, each asking five questions with relevance to this analysis. Respondents may show a response set bias on each prompt across sections or within each of the thirteen sections (rated all five responses to a common stimulus similarly). In other words, importance could have been ranked similarly for all thirteen of the functions. In the second case, in the first case each response in the section on staffing could be influenced by the response to the first item in the section which assesses the importance of that function to the organization. Each of these possibilities has been examined. To address the former

TABLE 2
Within-Subject Variances Across Mailback Questionnaire Items
N = 119

Items Measuring:	Descriptive Statistics for Subject Variances				
	Mean	Std Dev	Median	Minimum	Maximum
Importance	1.19	.79	1.02	.09	3.60
Complexity	1.08	.70	.96	.00	3.27
Rate of Change	1.02	.62	.87	.91	2.69
External Linkages	1.42	.88	1.22	.16	4.36
Internal Linkages	1.10	.82	.89	.00	3.47

concern, within-subject variances were computed for the parallel items across the thirteen sections.

The frequencies on these variance variables are presented in Table 2. Examination of these results suggest that the variances themselves are reasonably distributed. In addition, examination of the correlation matrices containing the intercorrelations of first the importance variables, then the complexity variables, and so on, indicates that the within-subject correlation across the thirteen sections is not of a problematic level. Granted, there are no decision guidelines which can be relied on in making these observations. Instead, the fact that the correlations in the matrices vary in strength, with stronger relationships between measures existing in sections that define a sector, is interpreted as an indication that an obvious problem does not exist. The second test concerning response set bias concerns a bias in responses within each of the 13 sections. To examine this possibility, correlations were computed among the variables in each section. Once again, it is difficult to ascertain any pattern of intercorrelation other than what would be expected, given the theoretical interrelationships. In addition, the patterns of intercorrelation vary by sector, an occurrence that would not be expected if a high degree of response set bias existed. While neither examination has revealed any obvious problem, it is important to note that since the variables are supposed to be correlated to an extent, it is hard to be certain that a problem with response set bias does not exist. For the present, it is argued that an obvious problem does not exist. Later analyses comparing the P/HR environmental sectors based on these thirteen sections of the questionnaire with other variables will be instrumental in reinforcing this assertion, but their presentation here would be premature.

In the second portion of the analysis, the three factor conceptualization of P/HR environmental uncertainty proposed above is examined using an exploratory factor analysis. This paper suggested that strategic uncertainty in the 13 P/HR areas measured

TABLE 3
 Factor Analysis for Eleven Uncertainty Measures
 Pattern Matrix
 Principal Axis Factoring with Oblique Rotation ^a
 N = 120

P/HR Activity	Factor 1	Factor 2	Factor 3
	Social	Task	Regulatory
Staffing	.045	.665	-.080
Compensation	.024	.628	.049
Performance Appraisal	.055	.533	.249
Government Regulations	-.078	-.023	.746
Employee Relations	.297	.242	.320
Safety	.530	-.143	.272
Alcohol Abuse	.948	-.050	-.052
Drug Abuse	.927	-.018	-.044
Family Problems	.540	.330	-.061
Health Care Cost Containment	.129	.116	.592
Healthy Lifestyle Issues	.665	.040	.000
Eigen Value	4.388	1.506	1.246
% Variance Explained	39.9	13.7	11.3

^a Items in bold represent loadings used to identify sectors.

could be theoretically considered as representative of three environmental sectors: social, task, and regulatory. In order to assess the adequacy of this statement, the strategic uncertainty measures in each of the 13 areas were subject to an exploratory factor analysis. Since it was noted that the sectors are expected to be correlated, an oblique rotation was used. The results of the first factor analysis (not reported) indicated that the AIDS variable cross loaded on the social and task sectors. Conceptually, AIDS represents a special case of a problem that not only does the workplace lack the technology to address, but society does, as well. Further, the determination of worksite responsibilities and employee rights has not been made. In this way, it is an even greater enigma for P/HR departments than the other social issues, for which some consensus on the technology with which they should be addressed exists. Given the recency of and urgency associated with AIDS in the workplace discussed above, it was decided that the AIDS issue would be considered independently of the three hypothesized sectors and is from this point on referred to as an emerging issue.

The factor analysis was repeated for strategic uncertainty in the twelve remaining areas of P/HR concern.¹ The pattern matrix from this factor analysis is presented in Table 3. Consideration of eigen values and the scree plot suggested the rotation of a three factor solution. This factor rotation yielded an interpretable factor pattern matrix with relatively simple structure and was retained.

As predicted, staffing, compensation, and performance appraisal all appear on a factor representing the P/HR task environment. The loadings on the remaining two factors are somewhat counter to what had been hypothesized and as such have a considerable bearing on the remainder of the analysis. First, the employee relations measure fails to load

¹ The analysis was also conducted for the uncertainty measure based on Duncan's model (C + R). The results do not differ appreciably. Later analysis will consider the utility of the Daft *et al* (1988) measure, the Duncan measure, and a third alternative measure.

TABLE 4
Descriptive Statistics for Sector Measures
N=119

Sector	Mean	Standard Deviation	Median	Chronbach's Alpha
<u>Task - Three Items</u>				
Importance	12.55	2.21	13.00	.60
Rate of Change	10.20	2.43	10.00	.62
Complexity	9.93	2.67	10.00	.65
Uncertainty	20.16	4.34	20.00	.65
Strategic Uncertainty	85.59	24.92	84.50	.61
External Linkages	8.05	2.82	8.00	.66
Internal Linkages	11.70	2.17	4.72	.78
<u>Social - Five Items</u>				
Importance	17.27	4.11	18.00	.79
Rate of Change	16.38	4.52	17.00	.83
Complexity	18.25	4.32	19.00	.83
Uncertainty	34.63	8.11	121.00	.86
Strategic Uncertainty	124.20	47.38	121.00	.83
External Linkages	16.36	5.05	18.00	.82
Internal Linkages	17.76	4.12	19.00	.78
<u>Regulatory - One Item</u>				
Importance	4.40	.90	5.00	
Rate of Change	3.81	1.14	4.00	
Complexity	3.84	1.23	4.00	
Uncertainty	7.64	2.04	4.18	
Strategic Uncertainty	34.36	12.47	35.00	
External Linkages	3.65	1.12	4.00	
Internal Linkages	3.93	1.04	4.00	
<u>Emerging Issue (AIDS) - One Item</u>				
Importance	2.97	1.21	3.00	
Rate of Change	3.08	1.38	3.00	
Complexity	4.35	1.05	5.00	
Uncertainty	7.42	1.97	8.00	
Strategic Uncertainty	22.73	12.01	24.00	
External Linkages	3.24	1.49	4.00	
Internal Linkages	3.28	1.41	4.00	

on any of the three factors. This has led to its exclusion for the remaining analyses. In addition, strategic uncertainty concerning safety loaded on the factor made up of social sector measures, as opposed to the regulatory sector factor. Post hoc, it is a relatively simple matter to develop an argument justifying the use of safety concerns as an indicator of the social sector. The regulatory sector argument would stress compliance with OSHA and avoidance of expensive worker's compensation claims, while a social sector argument would suggest that no organization wants its employees to be injured, but recognizes that factors over which it has no direct control, such as employee drug use, can have safety consequences. Based on this finding, the social sector was modified to include safety as an indicator. From this point, the analysis is focused on four P/HR environmental sectors based on the following sets of organizational concerns: (1) task, indicated by staffing, performance appraisal, compensation; (2) social, indicated by safety, alcohol abuse, drug abuse, family problems, healthy lifestyles; (3) regulatory, indicated by government regulations and health care cost containment; (4) emerging, indicated by AIDS.

The next step in the analysis was to compute measures of internal consistency for the scales from each of the sectors containing multiple indicators listed above (task, social, regulatory). Chronbach's alpha for these measures is presented in Table 4, along with descriptive statistics for all sector measures. When the alpha for the scales in the regulatory sector were examined, it became apparent that the use of health care cost containment in conjunction with government regulations was problematic due largely to the lack of variation in the former. Because the sector concern is more clearly identified by the single measure concerning regulations than by that addressing health care costs, only the former is considered from this point on.

Tests of Hypotheses

The next series of analyses were concerned with the tests of Hypotheses One through Four. Hypothesis One, which concerns the level of perceived uncertainty present in P/HR environmental sectors, is tested using a series of t-tests. The second hypothesis, which postulated a reciprocal relationship between external linkages and perceived strategic uncertainty, is tested using a two-stage least squares regression. Hypothesis Three concerned the presence of internal linkages and their relationship to external linkages was examined using zero order correlations. Finally, Hypothesis Four, which considers the organizational responses to a boundary spanner's interpretation of the environment, is tested using a stepwise ordinary least squares (OLS) regression.

CHAPTER IV

RESULTS

The results of the analyses are presented in two sections. The first section reports the findings from the tests of Hypotheses One through Four. Within this section, Hypothesis Two is considered last, because of a difficulty encountered in its test. The second section of the results presents a discussion of relative utility of three algorithms for measuring uncertainty.

Tests of the Hypotheses

Hypothesis One

Hypothesis One was based on the work by Daft et al (1988) on CEO scanning behavior and stated that sectors would differ in the amount of perceived uncertainty they contained, with sectors having the potential for the greatest impact on the organization's core technology having the greatest uncertainty. This hypothesis was tested using a series of pairwise t-tests. These results are presented in Table 5. As predicted, the regulatory environment contained the highest level of reported uncertainty. The first hypothesis suggested that the second highest amount of uncertainty would be found in the social sector. The analysis described above created the emerging issue category, which is a special case of a social issue. This emerging issue, AIDS, contained the next greatest amount of uncertainty, followed by social and finally task uncertainty. The difference between regulatory uncertainty and emerging uncertainty, however, was not significant. While the difference between social and task uncertainty was not significant, the means do rank the two in the predicted order. One further note is that the uncertainties, while having

TABLE 5
Difference in Means Tests (T-Tests) for Sector Strategic Uncertainty Indicators ^a
N = 115

Pair	Mean	Standard Deviation	Pearson Correlation	t
Regulatory Uncertainty &	7.65	2.06		
Emerging Uncertainty	7.42	1.97	.34***	1.08
Social Uncertainty	6.96	1.59	.31***	3.72***
Task Uncertainty	6.72	1.45	.16	4.69***
Emerging Uncertainty &	7.42	1.97		
Social Uncertainty	6.96	1.59	.58***	3.17**
Task Uncertainty	6.72	1.45	.43***	4.39***
Social Uncertainty &	6.96	1.59		
Task Uncertainty	6.72	1.45	.34***	1.42

* $p < .05$; ** $p < .01$; *** $p < .001$

^a Multiple item scales have been adjusted for the number of items to be on the same metric as single item measures.

mean differences, are still moderately correlated. This is to be expected, as the argument has been advanced herein that issues in each of the sectors can affect the others.

Hypothesis Three

This hypothesis suggested that departments which displayed strong external linkages would also be well connected within the organization, displaying strong internal linkages. It was further suggested that in sectors without consensual technologies to address uncertainty, the relationship would be stronger than in sectors with agreed upon methods for uncertainty reduction or coping. As predicted, in the task and regulatory sectors, the correlation between external and internal linkages is significant, positive, and weak ($r = .18$ and $.20$, respectively, with $p < .05$). In the social and emerging sectors, the correlations are quite strong ($r = .69$ and $.62$, respectively, with $p < .001$). These findings support Hypothesis Three.

Hypothesis Four

Hypothesis Four suggested that resources committed by the organization to address uncertainty in the social and task sectors would be greater in organizations where the P/HR department maintained strong external linkages. Table 6 presents the results from the stepwise OLS regression of resources committed on perceptions of strategic uncertainty, external linkages, and internal linkages. Since the resources measure focuses on social and task issues, relationships were not expected in the regulatory sector. Of interest is the third step of the regression, which considers the role of external linkages on resources committed while controlling for perceptions of uncertainty and internal linkages. As expected, there is no relationship between external linkages in the regulatory sector and resources committed by the organization. In the task, social, and emerging sectors, however, a strong positive relationship was found between external linkages and resources committed, supporting Hypothesis Four.

TABLE 6
Results from Multiple Regressions of Resources Committed on
External Linkages, Internal Linkages, and Perceived Strategic Uncertainty

Independent Variable and Sector	Dependent Variable Resources Committed		
	<i>Equation 1</i>	<i>Equation 2</i>	<i>Equation 3</i>
Task (N = 119)^a			
Strategic Uncertainty	.18*	.08	.07
External Linkages		.36***	.35***
Internal Linkages			.05
R ²	.03	.15	.15
Regulatory (N = 123)			
Strategic Uncertainty	-.04	-.11	-.12
External Linkages		.17	.16
Internal Linkages			.10
R ²	.00	.03	.04
Social (N = 115)			
Strategic Uncertainty	.14	-.16	-.13
External Linkages		.60***	.64***
Internal Linkages			-.09
R ²	.02	.26	.28
Emerging (N = 119)			
Strategic Uncertainty	.16	-.13	-.17
External Linkages		.56***	.52***
Internal Linkages			.09
R ²	.02	.25	.26

^a N reported is based on a pairwise deletion.

Hypothesis Two

Hypothesis 2 addresses the relationship between the external linkage component of boundary spanning and perceptions of uncertainty. The hypothesis suggests that in institutionalized sectors, boundary spanning occurs as a matter of routine based on technologies the organization has developed to address uncertainty. Alternatively, the relationship was expected to be reciprocal in sectors that have not been institutionalized and as such do not have well developed or consensual technologies to address them. In this analysis, no relationship was expected between internal linkages and perceived strategic uncertainty in the task and regulatory sectors, while a positive reciprocal relationship was expected in the social and emerging sectors.

In 2SLS, exogenous (independent) variables are used to partial out the mutual influence of two dependent variables. 2SLS is one of a family of methods that is typically considered "confirmatory". In a confirmatory analysis, the researcher is able, based on a theoretical foundation, to make stronger inferences regarding causality than in a non-confirmatory (exploratory) analysis (James, Mulaik, and Brett, 1982). James, *et al* (1982) outline ten conditions which must be met in order to consider a confirmatory analysis including: the formal statement of theory in terms of a structural model, theoretical rationale for causal hypotheses, specification of causal order and direction, self-contained equations (all relevant causes of the dependent variables are measured) and the specification of the conditions, or boundaries, under which the hypothesized relationships hold (p 26).

The theoretical rationale for the relationship between external linkages and perceptions of uncertainty has been developed in this paper, and is well documented in the literature. This portion of the model portrayed in Figure 1 can be justifiably tested using a confirmatory procedure, such as 2SLS. There is, however, no literature which supports the consideration of exogenous variables which would be used to estimate the reciprocal

relationship between external linkages and uncertainty. In terms of the conditions of confirmatory analysis noted briefly above, the reciprocal relationship is based on theoretical rationale, specifies causal order and direction, and specifies the conditions under which the relationship should hold (e.g. institutionalized versus "new" sectors). However, there is at this point in time no way to develop self contained equations including all relevant causes of the dependent variables. As a result, there is no "model" which can be confirmed or disconfirmed through this analysis, indicating that it would be premature to attempt a confirmatory analysis strategy.

Despite the lack of theoretically justified exogenous variables to consider, it is argued here that in order to examine the existence of the reciprocal relationship between external linkages and uncertainty, exogenous variables from a family of variables which have been shown in previous research to be important to organizational theory models (e.g. organizational size) can be selected and used to specify a 2SLS model. In essence, the analysis picks independent variables that are theoretically plausible, but untested, to examine the relationship of interest - that between external linkages and perceived strategic uncertainty.¹ The important point to emphasize here is that 2SLS in this analysis is being used as an exploratory technique. While this approach is feasible, there are difficulties putting it into practice, as will be summarized below.

In order to select the independent variables used in the 2SLS analysis, a bivariate correlation matrix of external linkages, perceived uncertainty, and 25 variables from the interview and the mailback questionnaire was developed for each of the four sectors. The correlation matrix was examined and variables which were strongly correlated with either

¹ This is similar to the use of a computer generated independent variable that can be used in LISREL models to specify an equation. One advantage here is that the independent variables, while still somewhat arbitrary, are not devoid of conceptual relevance, as a computer generated variable would be.

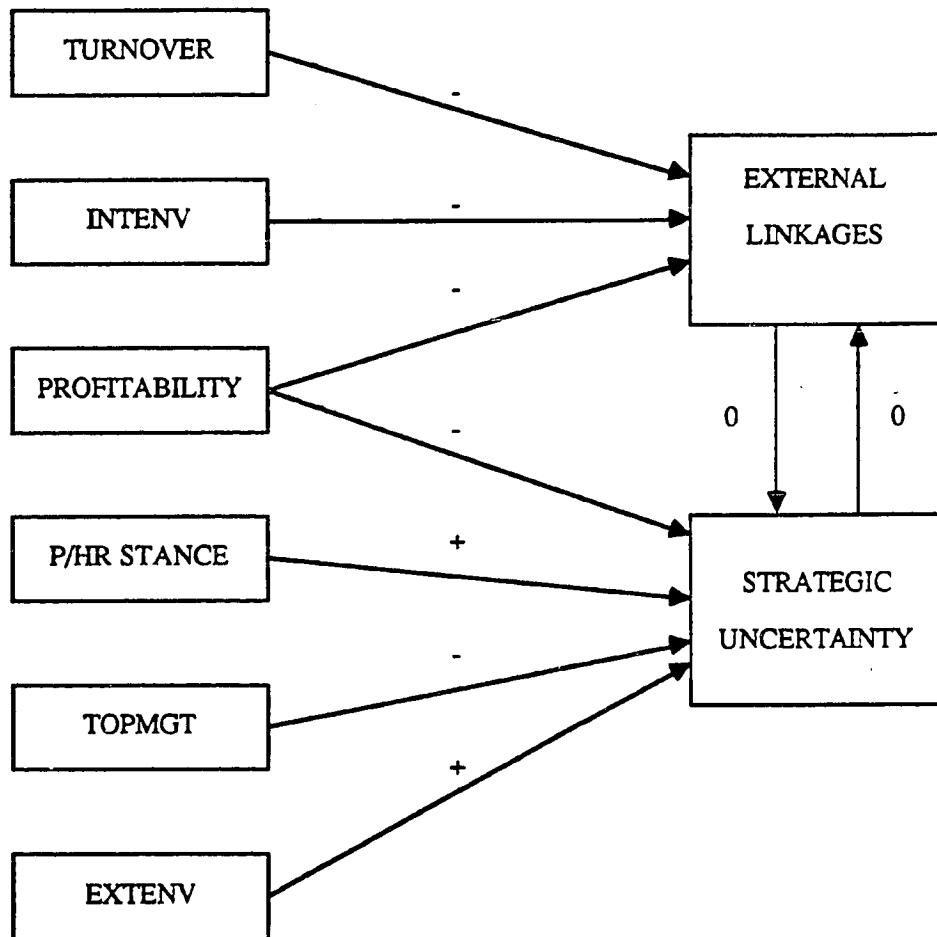


FIGURE 2

**Model Relating External Linkages and Perceptions of
Strategic Uncertainty to Exogenous Variables
TASK ENVIRONMENT**

TABLE 7

Pearson Correlations Between Predicted and Observed Dependent Variables
and Indicator Variables in the Task Sector ^a

	1	2	3	4	5	6	7	8	9	10
1. External Linkages	1.00									
2. Strategic Uncertainty	.28	1.00								
3. Predicted External Linkages	.34	.48	1.00							
4. Predicted Strategic Uncertainty	.30	.56	.83	1.00						
5. Turnover	-.22	-.12	-.65	-.30	1.00					
6. INTENV	-.17	-.11	-.49	-.26	.16	1.00				
7. P/HR Stance	.14	.41	.51	.78	-.11	-.04	1.00			
8. Company Profitability	-.11	-.13	-.28	-.23	-.01	.13	.09	1.00		
9. TOPMGT	-.09	-.28	-.27	-.51	-.06	-.03	-.23	.16	1.00	
10. EXTENV	.07	.21	.21	.28	-.09	.16	.12	.01	-.21	1.00

^a A correlation of .17 is significant with $p < .05$, of .23 with $p < .01$, or .28 with $p < .001$.

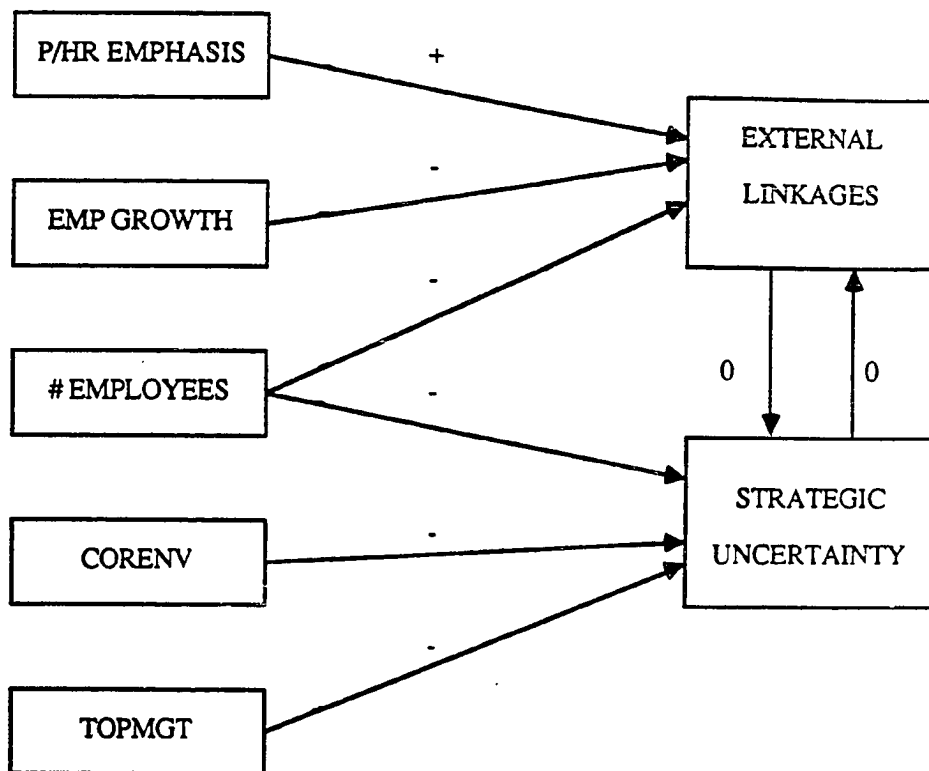


FIGURE 3
Model Relating External Linkages and Perceptions of
Strategic Uncertainty to Exogenous Variables
REGULATORY ENVIRONMENT

TABLE 8
Pearson Correlations Between Predicted and Observed Dependent Variables
and Indicator Variables in the Regulatory Sector ^a

	1	2	3	4	5	6	7	8	9
1. External Linkages	1.00								
2. Strategic Uncertainty	.40	1.00							
3. Predicted External Linkages	.31	.20	1.00						
4. Predicted Strategic Uncertainty	.22	.34	.73	1.00					
5. P/HR Emphasis	.21	.13	.65	.39	1.00				
6. Employee Growth/Loss	-.17	-.08	-.55	-.29	-.15	1.00			
7. Number of Employees	-.16	-.15	-.51	-.48	.16	.04	1.00		
8. CORENV	-.02	-.15	-.11	-.45	-.08	.02	.18	1.00	
9. TOPMGT	-.10	-.25	-.32	-.75	-.12	.08	.15	.06	1.00

^a A correlation of .15 is significant with $p < .05$, of .21 with $p < .01$, or .30 with $p < .001$.

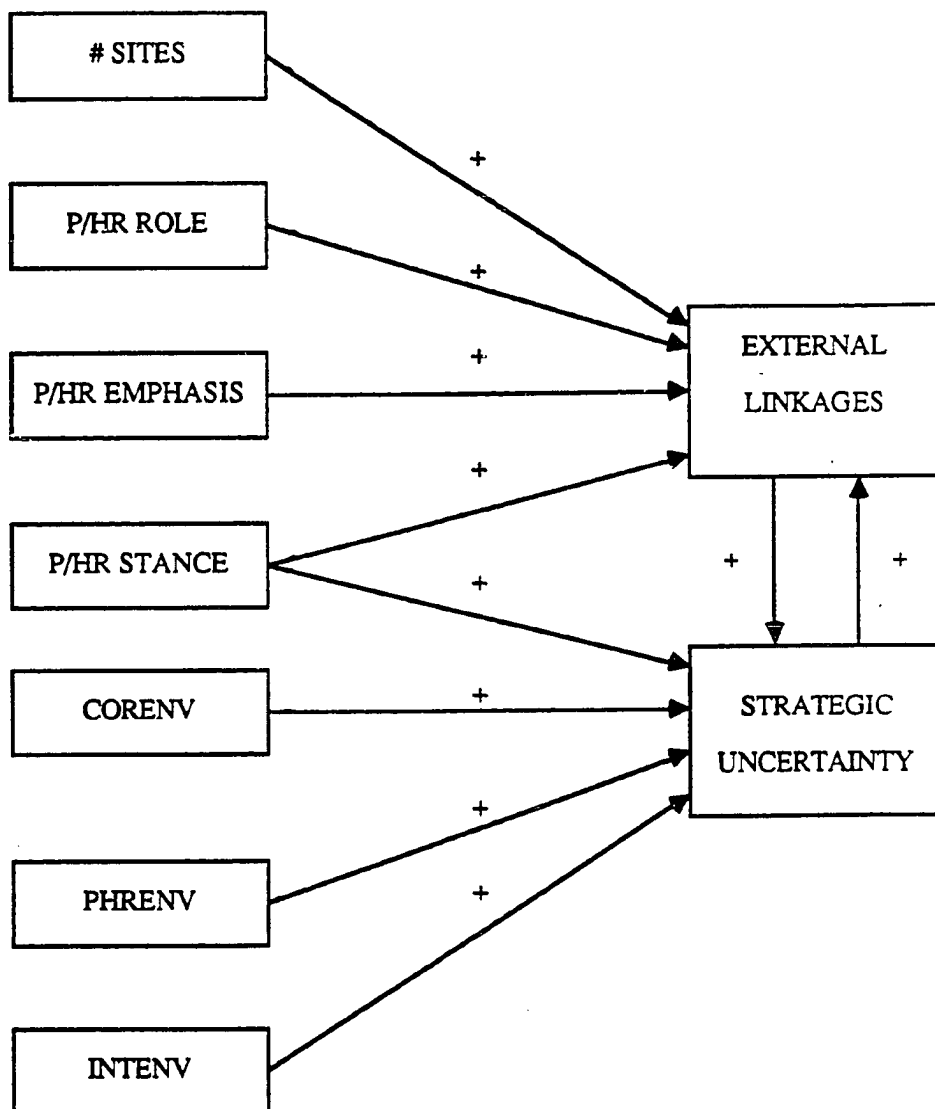


FIGURE 4

Model Relating External Linkages and Perceptions of
Strategic Uncertainty to Exogenous Variables
SOCIAL ENVIRONMENT

TABLE 9

Pearson Correlations Between Predicted and Observed Dependent Variables
and Indicator Variables in the Social Sector ^a

	1	2	3	4	5	6	7	8	9	10	11
1. External Linkages	1.00										
2. Strategic Uncertainty	.51	1.00									
3. Predicted External Linkages	.44	.16	1.00								
4. Predicted Strategic Uncertainty	.15	.53	.29	1.00							
5. N of Sites	.30	.02	.69	-.01	1.00						
6. P/HR Role	.30	.03	.72	.07	.30	1.00					
7. P/HR Emphasis	.22	.10	.52	.19	.23	.37	1.00				
8. P/HR Stance	.14	.20	.26	.37	.13	.18	.45	1.00			
9. CORENV	.05	.15	.10	.29	.18	.03	-.08	.05	1.00		
10. PHRENV	.05	.22	.15	.42	-.11	-.11	-.16	-.11	-.05	1.00	
11. INTENV	-.07	.41	-.14	.78	-.16	-.21	-.30	-.04	.01	.39	1.00

^a A correlation of .15 is significant with $p < .05$, of .22 with $p < .01$, or .29 with $p < .001$.

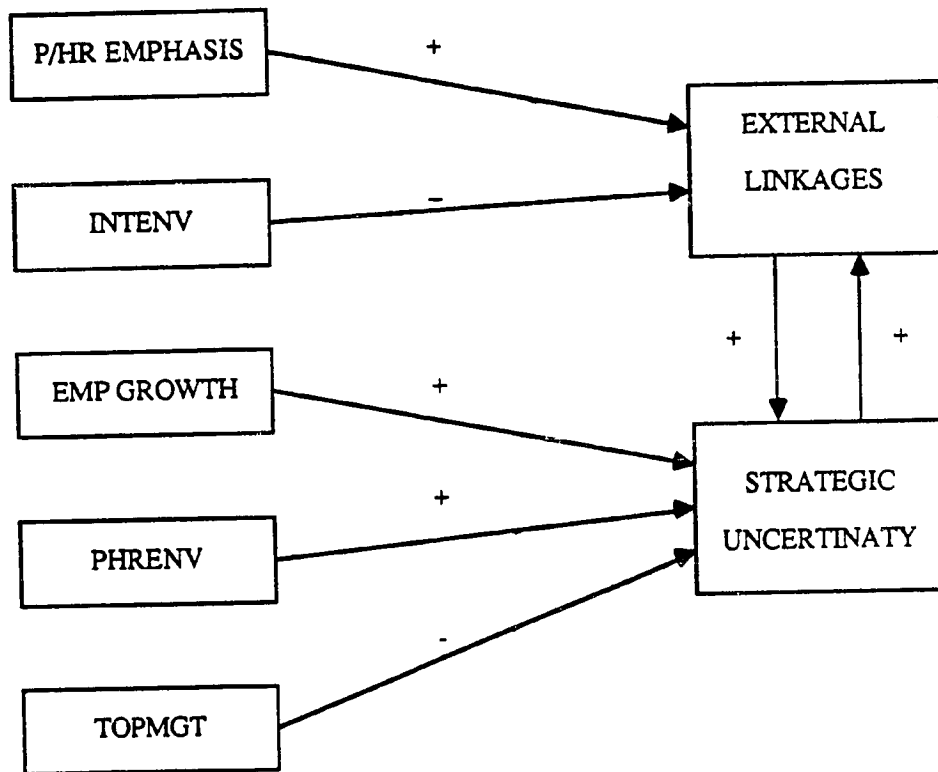


FIGURE 5

Model Relating External Linkages and Perceptions of
Strategic Uncertainty to Exogenous Variables
EMERGING ENVIRONMENT

TABLE 10
Pearson Correlations Between Predicted and Observed Dependent Variables
and Indicator Variables in an Emerging Area (AIDS) ^a

	1	2	3	4	5	6	7	8	9
1. External Linkages	1.00								
2. Strategic Uncertainty	.51	1.00							
3. Predicted External Linkages	.38	.16	1.00						
4. Predicted Strategic Uncertainty	.19	.31	.51	1.00					
5. P/HR Emphasis	.23	.13	.71	.41	1.00				
6. INTENV	-.26	-.01	-.79	-.02	-.30	1.00			
7. Employee Growth/Loss	.06	.14	.22	.66	.11	.11	1.00		
8. PHRENV	-.07	.20	-.22	.61	-.16	.39	.04	1.00	
9. TOPMGT	-.13	-.22	-.37	-.68	-.12	-.03	-.20	-.17	1.00

^a A correlation of .16 is significant with $p < .05$, of .22 with $p < .01$, or .29 with $p < .001$.

external linkages, perceived strategic uncertainty, or both were selected. These variables became the independent variables presented in Figures 2 through 5. It is significant to note that a number of variables which are consistently important in organizational theory and personnel management models were not correlated with either dependent variable. For example, organizational size, union representation, and whether the site was part of a corporate structure or an independent, single site company were not correlated significantly with either internal linkages or uncertainty.

Once developed, the models in Figures 2 through 4 were analyzed using a 2SLS procedure. The first stage of the two-stage procedure is to regress each of the dependent variables on all of the independent variables in the model, saving the predicted values of the dependent variable. These predicted values are free from correlation with the unmeasured causes of the variable with which they are reciprocally related (James *et al.* 1982). This procedure is repeated for each dependent variable in each sector, resulting in eight regression runs. Before moving on to the second stage of the procedure, the correlation matrix between the dependent variables, the new predicted dependent variables, and the independent variables are examined. These correlation matrices are presented in Tables 7 through 10. When looking at the matrices, several things should appear which would justify moving on to the second stage of the analysis. In each of the four Tables, the three most important correlations are those between (1) the predicted values of the dependent variables, and (2) the two correlations between each predicted value and its original value. In order to continue the analysis, the first correlation should be quite high. There are no standards for just how high this correlation needs to be, but convention seems to suggest it should be .80 or better. This is true only in the task sector matrix presented in Table 7. The correlations between each predicted value of the dependent variable and its original value should also be high, though not necessarily as high as the correlation between the

two predicted values. In other words, the model should account for considerable variance in the dependent variable. Again, there are no guidelines for the acceptable range of this correlation, but the values reported in Tables 7 through 10, which range from .31 to .53, are clearly not strong enough to warrant continuation of the 2SLS procedure. Essentially, the weakness of these correlations indicates that the variables chosen as independent variables are not adequate as instruments with which to disentangle the direction of the relationship between the two dependent variables.

Given the failure of the data to support a two-stage test of the relationship between external linkages and perceived uncertainty, the analysis was reorientated towards an ordinary least squares (OLS) approach. Obviously, the OLS method does not provide a way to test the reciprocal relationship between the dependent variables, so Hypothesis Two must remain untested. Instead, the OLS analysis is used to assess the robustness of the correlation between external linkages and perceived uncertainty. In each sector, regression analyses were conducted using alternatively external linkages and strategic uncertainty as dependent variables. Independent variables included the set of variables which had been most consistently important in the aborted 2SLS procedure outlined above (P/HR Role, P/HR Emphasis, P/HR Stance, CORENV, EXTENV, INTENV, and PHRENV). Additionally, the set of traditional organizational theory control variables which did not appear, based on bivariate correlations, to have bearing on boundary spanning or perceived uncertainty (number of employees, turnover, union presence/absence, company profitability, and relationship of location to corporate headquarters) were included to see if consideration of the multivariate relationships would indicate whether these variables affected the external linkage - perceived uncertainty relationship. In each regression, the dependent variable (either external linkages or perceived uncertainty) was regressed on the set of independent variables which were entered into the regression in a stepwise fashion.

Once all the variables which met the tolerance requirements had been entered, a final independent variable (either perceived uncertainty or external linkages) was entered. The results from these analyses are presented in Tables 11 through 14.

From these regressions, two findings are of interest. First, in the institutionalized sectors of the P/HR environment (task and regulatory), the only variable that explained perceptions of uncertainty was the measure of external linkages. In contrast, in the social and emerging sectors, several variables contributed to an understanding of perceived uncertainty. In the model for the social sector, the Betas for P/HR role and external linkages were significant, explaining 35% of the variance in perceived uncertainty. Departments that reported having a consultative role in the organization were more likely, as were those who displayed higher levels of external linkages, to report higher degrees of perceived uncertainty regarding issues in the social sector (P/HR role and external linkages are correlated .30 with $p < .001$). In the emerging sector, P/HR role remains an important predictor of perceived and top management concerns about the external social climate are initially contributors to variance explained in strategic uncertainty, but when the effects of external linkages are considered, the relationships are no longer significant. It is interesting that concern about the internal employment environment (INTENV) is negatively associated with perceived uncertainty in the emerging sector. In other words, where top management was less concerned with issues like absenteeism, P/HR departments reported higher levels of uncertainty present regarding AIDS. Overall, INTENV and external linkages explained 37% of the variance in perceived uncertainty.

The results of the regressions with external linkages as the dependent variable are less clear, indicating that different independent variables were important in each of the four sectors. P/HR stance is a predictor in the task, regulatory, and emerging sectors, although in the latter the entry of perceived uncertainty eliminates its significance. In each case,

TABLE 11
 Betas from Multiple Regression of External Linkages and
 Strategic Uncertainty on Independent Variables

Task Sector

N = 96 ^a

Dependent Variables

Independent Variables	External Linkages			Strategic Uncertainty
	Step 1	Step 2	Step 3	Step 1
# Employees				
Turnover (log)				
Union Presence (1 = union)				
Profitability				
Relationship to Corporate				
P/HR Stance		.28*	.26*	
P/HR Role				
P/HR Emphasis	.41***	.28*	.27*	
CORENV				
EXTENV				
INTENV				
PHRENV				
External Linkages				.28*
Strategic Uncertainty			.21*	
R ²	.17	.23	.28	.08

^a Minimum pairwise N

* p < .05; ** p < .01; *** p < .001

TABLE 12
 Betas from Multiple Regression of External Linkages
 and Strategic Uncertainty on Independent Variables

Regulatory Sector

N = 104 ^a

Dependent Variables

Independent Variables	External Linkages			Strategic Uncertainty
	Step 1	Step 2	Step 3	Step 1
# Employees				
Turnover (log)				
Union Presence (1 = union)				
Profitability				
Relationship to Corporate				
P/HR Stance	.34**	.35*	.29*	
P/HR Role				
P/HR Emphasis				
CORENV				
EXTENV				
INTENV		.25*	.19	
PHRENV				
External Linkages				.40***
Strategic Uncertainty			.31*	
R ²	.12	.18	.27	.16

^a Minimum pairwise N

* p < .05; ** p < .01; *** p < .001

TABLE 13
 Betas from Multiple Regression of External Linkages
 and Strategic Uncertainty on Independent Variables

Social Sector

N = 101 ^a

Dependent Variables

Independent Variables	External Linkages		Strategic Uncertainty	
	Step 1	Step 2	Step 1	Step 2
# Employees				
Turnover (log)				
Union Presence (1 = union)				
Profitability				
Relationship to Corporate				
P/HR Stance				
P/HR Role			.30*	.29**
P/HR Emphasis				
CORENV				
EXTENV				
INTENV	.41***	.45***		
PHRENV				
External Linkages				.50***
Strategic Uncertainty		.54***		
R ²	.17	.46	.09	.35

^a Minimum pairwise N

* p < .05; ** p < .01; *** p < .001

TABLE 14
 Betas from Multiple Regression of External Linkages
 and Strategic Uncertainty on Independent Variables
 Emerging Sector
 N = 106 ^a

Independent Variables	External Linkages			Strategic Uncertainty			
	Step 1	Step 2	Step 3	Step 1	Step 2	Step 3	Step 4
# Employees							
Turnover (log)							
Union Presence (1 = union)							
Profitability							
Relationship to Corporate							
P/HR Stance		.26*	.19				
P/HR Role				.32*	.31*	.25*	.17
P/HR Emphasis							
CORENV							
EXTENV	.37**	.34**	.25*		.25*	.29*	.14
INTENV						-.26*	-.25*
PHRENV							
External Linkages							.42***
Strategic Uncertainty			.40***				
R ²	.14	.21	.35	.10	.16	.22	.37

^a Minimum pairwise N

* p < .05; ** p < .01; *** p < .001

departments rated by the interviewer as proactive reported greater external linkages. While the other variables form an inconsistent pattern across sectors, the significant relationships make sense conceptually.

In the regulatory and social sectors, concern with the employment issues such as absenteeism, turnover, and workers compensation (INTENV) is an important predictor of the existence of external linkages. Certainly, organizations with these concerns would be expected to show stronger links with external constituents on issues concerning regulations. Additionally, many of the social sector issues, like alcohol and drug abuse, have bearing on absenteeism, turnover, and workers compensation, and as such could be addressed through the maintenance of external linkages. In the emerging sector, concern with the external environment (EXTENV) is important. As noted earlier, organizations have no technology by which to address AIDS. Further, the legal consequences of the way AIDS is handled is unclear, but potentially dramatic. P/HR departments which monitor the external environment would be expected to have a greater understanding of these possibilities.

Finally, in the task sector, P/HR emphasis and P/HR stance are useful predictors of external linkages. The results in Table 11 suggest that departments classified by the interviewer as proactive and as having an emphasis on innovative P/HR practice ($r = .45$; $p < .001$) maintain more active external linkages. In this institutionalized sector of the P/HR environment this paper has argued that external links are not necessary. The fact that departments concerned with innovation maintain links despite low uncertainty would suggest that these organizations would be the early adopters of new P/HR technologies. In all cases, perceived uncertainty explained further variance in external linkages, supporting Thompson's theory. It is remarkable that in the multivariate model, the set of traditional measures failed to explain variance in the dependent variables, yet the external linkage -

perceived uncertainty association is consistent. One explanation for this finding is that the external linkage - perceived uncertainty relationship is robust across the variables considered. A second explanation is that by focusing on the Atlanta SMSA, the analysis has truncated variance in these traditional measures. While an analysis of the variable distributions does not suggest this is a problem, it remains an issue for future analyses.

Other Analyses

Complexity and Rate of Change

In his discussion of environmental uncertainty, Duncan (1972) develops a two by two table with dimensions labelled complexity and rate of change. Each dimension was dichotomized into a high and low category. Uncertainty was said to be least in the "low-low" condition and greatest in the "high - high" condition. The degree to which uncertainty was a problem in the other conditions ("high - low"; "low - high") was less clear. Duncan argued that of the two dimensions, rate of change was the most problematic for organizations. As Thompson (1967) suggested, organizations in a complex environment can develop structures that are isomorphic with the environment, or use strategies such as stockpiling, to address complexity. Organizations are less suited, the argument continues, to address a high rate of change. This view is shared not only by early organizational theorists, but in more recent theories, such as population ecology (Hannan and Freeman, 1977). This would lead one to suspect that boundary spanning strategies would be more strongly related to perceptions of a rapidly changing environment than to perceptions of environmental complexity. While the differences in correlations between boundary spanning strategies and complexity and boundary spanning strategies and rate of change are not significantly different statistically, 7 of the 8 pairs in Tables 15 through 18 show a

TABLE 15
Pearson Correlation Coefficients: Items in the P/HR Task Sector a

N = 123

Variable	1	2	3	4	5	6	7	8
1. Importance	1.00							
2. Complexity	.02	1.00						
3. Rate of Change	.16	.44	1.00					
4. Uncertainty (C + R)	.10	.87	.83	1.00				
5. Strategic Uncertainty ((D)(C + R))	.64	.67	.71	.81	1.00			
6. Additive Uncertainty (I + C + R)	.53	.75	.78	.90	.98	1.00		
7. External Linkages	.15	.21	.26	.28	.28	.31	1.00	
8. Internal Linkages	.39	.02	.15	.10	.27	.25	.18	1.00

a $r > .15$ is significant with $p < .05$; $r > .24$ with $p < .01$; $r > .27$ with $p < .001$

TABLE 16
Pearson Correlation Coefficients: Items in the P/HR Regulatory Sector ^a

N = 123

Variable	1	2	3	4	5	6	7	8
1. Importance	1.00							
2. Complexity	.30	1.00						
3. Rate of Change	.45	.49	1.00					
4. Uncertainty (C + R)	.43	.87	.85	1.00				
5. Strategic Uncertainty (I)(C + R))	.75	.75	.80	.90	1.00			
6. Additive Uncertainty (I + C + R)	.70	.75	.83	.94	.98	1.00		
7. External Linkages	.35	.26	.38	.37	.40	.42	1.00	
8. Internal Linkages	.23	.10	.04	.09	.18	.16	.20	1.00

a $r > .16$ is significant with $p < .05$; $r > .23$ with $p < .01$; $r > .30$ with $p < .001$

TABLE 17
Pearson Correlation Coefficients: Items in the P/HR Social Sector^a

N = 123

Variable	1	2	3	4	5	6	7	8
1. Importance	1.00							
2. Complexity	.35	1.00						
3. Rate of Change	.50	.69	1.00					
4. Uncertainty (C + R)	.46	.92	.92	1.00				
5. Strategic Uncertainty ((I/(C + R))	.83	.73	.83	.85	1.00			
6. Additive Uncertainty (I + C + R)	.74	.83	.89	.94	.97	1.00		
7. External Linkages	.45	.38	.45	.45	.51	.52	1.00	
8. Internal Linkages	.55	.51	.57	.58	.65	.66	.69	1.00

^a all correlations reported are significant with $p < .001$

TABLE 18
Pearson Correlation Coefficients: Items in an Emerging Sector (AIDS) ^a
N = 123

Variable	1	2	3	4	5	6	7	8
1. Importance	1.00							
2. Complexity	.16	1.00						
3. Rate of Change	.36	.29	1.00					
4. Uncertainty (C + R)	.34	.74	.86	1.00				
5. Strategic Uncertainty (I)(C + R))	.88	.42	.65	.68	1.00			
6. Additive Uncertainty (I + C + R)	.72	.63	.80	.90	.92	1.00		
7. External Linkages	.45	.23	.43	.42	.51	.52	1.00	
8. Internal Linkages	.55	.22	.46	.44	.58	.57	.62	1.00

a $r > .16$ is significant with $p < .05$; $r > .22$ with $p < .01$; $r > .29$ with $p < .001$

stronger correlation between boundary spanning strategies and rate of change in the environment.

The point was raised earlier that organizations have less fully developed mechanisms, or technologies, to address uncertainties in the social and emerging sectors. The regulatory and task sectors, which formed the basis for the development of the personnel function from the early 1900s through World War II are suggested to have institutionalized technologies to address uncertainty. At this point, it is interesting to note that the correlations between external linkages and perceived strategic uncertainty, while not statistically different, are greater in the social and emerging sectors (.51, $p < .001$ in each case) than in the task and regulatory sectors (.28, $p < .01$ and .40, $P < .001$ respectively). This same pattern holds if one examines the correlation between external linkages and the other two algorithms for computing uncertainty.

Algorithms for Measuring Uncertainty

This analysis earlier alluded to a number of alternative algorithms for measuring perceived uncertainty. For the analysis, the Daft *et al* (1988) formulation was considered for theoretical reasons. A parallel series of analyses were conducted using two alternative formulations of uncertainty. The results of these analyses are of interest, because while the Daft *et al* (1988) measure is conceptually acceptable, no evidence was presented by Daft *et al* (1988) of any empirical examination of their measure or others.

Tables 15 through 18 present intercorrelations for a number of variables within each sector. Each table includes the importance, complexity, and rate of change used to describe an area by the respondents. In addition, the measures of boundary spanning (external linkages and internal linkages) are included. Finally, three measures of uncertainty are presented. To this point, the analyses presented have used the Daft *et al* (1988) measure of strategic uncertainty, although an unreported analysis was conducted simultaneously on the

other formulations presented in Tables 15 through 18. Daft *et al* (1988) argued that the measure of uncertainty based strictly on Duncan's (1972) work was not appropriate to the study of scanning behavior because even if an area contained high degrees of uncertainty, no scanning would occur in areas of little importance to the organization. As a result, they argue that Duncan's formulation of uncertainty ($C + R$) be weighted by the importance of an area, resulting in a measure of strategic uncertainty ($(I)(C + R)$). There is no conceptual support for the conclusion that a multiplicative function is appropriate, nor any indication that other algorithms were considered. Since one straightforward and parsimonious model was omitted in the move to a multiplicative function, Tables 15 through 18 also report correlations for the additive model ($I + C + R$). In addition to examining the uncertainty formulations, the discussion of the correlation matrices will focus on: (1) the relationship between complexity, rate of change, and boundary spanning strategies, and (2) the relationship between uncertainty and boundary spanning strategies.

Generally speaking, the conceptual rationale behind the Daft *et al* (1988) measure of strategic uncertainty is appealing. However, rather than accept the Daft *et al* (1988) measure of strategic uncertainty uncritically, several analyses were undertaken to empirically assess the appropriateness of the measure. Examining the correlations between the three uncertainty measures reported in Tables 15 through 18 indicates an extremely high pattern of intercorrelation. The only correlation less than .80 occurs in the emerging sector between the measure of strategic uncertainty and uncertainty ($r = .68, p < .001$). Essentially, this indicates that the importance component in the emerging sector had greater variance than in other sectors. From the consideration of the intercorrelations, it appears that the three uncertainty measures are essentially the same empirically. This was examined in greater detail through a regression analysis of boundary spanning on the measures of uncertainty. In each sector, boundary spanning was regressed on the uncertainty ($C + R$)

TABLE 19

Betas From Multiple Regressions of Strategic Uncertainty Measures on
Reports of External Linkages

N = 123

Independent Variable	Dependent Variable - P/HR Sector External Linkages							
	Regulatory		Social		Task		Emerging	
	Eq. 1	Eq. 2	Eq. 1	Eq. 2	Eq. 1	Eq. 2	Eq. 1	Eq. 2
	Eq. 1	Eq. 2	Eq. 1	Eq. 2	Eq. 1	Eq. 2	Eq. 1	Eq. 2
Uncertainty (C + R)	.27**		.32***		.28***		.30***	
Importance (I)	.25**		.31***		.11		.36***	
Additive Uncertainty (I + C + R)	.43***		.53***		.31***		.52***	
Strategic Uncertainty (I)(C + R)	.42***		.52***					.52***
R ²	.19	.19	.28	.28	.10	.10	.29	.27
		.18		.27		.08		.28

measure and importance, the additive model of uncertainty, and the Daft et al (1988) measure of strategic uncertainty. The results of these regressions are presented in Table 19. As expected, the regression indicates that when the three measures are used to predict external linkages, the additive and strategic measures are essentially identical.

CHAPTER V

CONCLUSIONS

Summary of Results

This dissertation began with a discussion of the increasingly important role that an organization's environment has been given in the development of organizational theory. In examining how an organization can address environmental uncertainty, thereby approximating the norms of rationality desired by organizational members, Thompson (1967) suggested that organizations create boundary spanning units which are charged by the organization with the tasks of monitoring, influencing, interpreting, and recounting important portions of the environment. A review of the research that has considered Thompson's boundary spanning concept found that most attention had been focused on the consequences faced by individuals in roles with boundary spanning activity.

This emphasis presents two problems. The first, concerning the difference between boundary spanning activity and the position of a boundary spanner, was addressed theoretically. The argument was made that boundary spanning activity, generally defined in previous empirical studies as any interaction with any environmental component, was unrelated to Thompson's conceptualization of boundary spanning. In essence, boundary spanning activity is merely one activity engaged in by a person with boundary responsibility. This confounding of definitions makes the use of Thompson's boundary spanning term difficult, so the distinction made by Tushman and Scanlan (1981a, b) between internal and external linkages is important. The research on boundary spanning to date has been concerned with individuals who have external linkages. Thompson's conceptualization posed that boundary spanners had not only external linkages, but internal

linkages, as well. It is not enough that an individual be exposed to the environment; to be a boundary spanner that individual must be actively disseminating the information within the organization in an effort to reduce uncertainty. As a result of this definitional quandary, the research on "boundary spanning" (here relabelled "external linkages") does not provide an adequate test for the hypotheses developed concerning the consequences of being a boundary spanner.

The second difficulty with the research on boundary spanning has been a near complete omission of the investigation of the relationship between boundary spanning and perceptions of uncertainty. Two reasons are offered to explain the dearth of research on this, the main tenet of Thompson's theory. First, it is possible that on the surface, the relationship has such face validity that the assumption remained untested. Second, it is suggested that difficulties involved in testing the relationship because of its instability over time have discouraged investigators. Essentially, characteristics of the environment and the organization's history in addressing that component of the environment can lead the relationship to be either reciprocal and positive or absent altogether. Methodologically, this presents significant problems to a consideration of the relationship.

The analysis presented here was designed to address these weaknesses. The dissertation reviewed the forces behind the establishment and development of the P/HR function, revealing that from the outset, forces in the organization's environment, such as the federal government, created an uncertainty that the organization could not address. As Thompson would suggest, organizations modified their structure to match that of the environment by developing a personnel practice to address these issues, thereby reducing uncertainty that could have had a detrimental impact on the organization's functioning. The expansion of the P/HR role in organizations was also considered to be the result of an

organizational need to address new contingencies in the personnel/human resource environment.

The analysis considered three hypotheses, the first of which was based on work by Daft *et al.* (1988) on environmental scanning. Sectors of the P/HR environment were identified and grouped based on the results of a series of factor analyses. The resulting four sectors, task, regulatory, social, and emerging, were compared in terms of the amount of uncertainty present and their proximity to the core of the organization. The first hypothesis suggested that areas with greater importance to the core of the organization would contain the greatest uncertainty. While not all the mean differences were significantly different, the uncertainties were in the rank order suggested by the hypothesis.

The second hypothesis concerned the relationship between the external linkage component of boundary spanning and perceptions of environmental uncertainty. As mentioned, this relationship is expected to be different, depending on characteristics of the environmental sector under consideration. That is, in areas where organizations have an agreed upon method for addressing uncertainty, no relationship between boundary spanning and uncertainty is expected. In areas characterized by a lack of consensus concerning the appropriate technology to employ in addressing uncertainty, however, a positive and reciprocal relationship is expected. This argument is based not only on Thompson's work, but that by institutional theorists (e.g. Tolbert and Zucker, 1983). While the relationship between boundary spanning and uncertainty is well documented theoretically, there has been no attention towards and therefore no understanding of the other variables which might be important to the relationship. Unfortunately, to empirically test the reciprocal relationship between two variables, other instrumental variables are needed. The analysis examined a series of variables which are generally important to models in organizational theory (e.g. size), but was unable to develop a model that could be

empirically tested. As a result, hypothesis two could not be tested. Instead, a series of OLS regressions were conducted which illustrated the robustness of the external linkage - perceived uncertainty relationship in a multivariate framework that controlled for a number of organizational and departmental characteristics.

Hypothesis Three examined the relationship between the internal and external linkages displayed by P/HR managers. The results supported the distinction made by Thompson (1967) and Tushman and Scanlan (1981a, b), in that both were present in areas of high uncertainty. Further, Hypothesis Four received support, indicating that organizational resources designated to address uncertainty were more likely to be found in organizations where P/HR departments were active as boundary spanners through both external and internal linkages.

Limitations and Suggestions for Future Research

There are a number of limitations to the present analysis which need to be discussed. The most obvious difficulty encountered in the analysis was the inability to specify a series of models with which to test Hypothesis Two. While the previous state of theory on which to base model development was weak, this is still a disappointment. Future research that is concerned with boundary spanning has two choices based on the findings of this analysis; the relationship between boundary spanning and perceived uncertainty can be taken for granted or efforts can be made to develop a better methodology with which to consider it. It is suggested that the prudent choice is the latter one. In attempting to construct a better test of the hypothesis, a consideration of this study's other limitations is helpful.

There are three related problems concerning the data analyzed in this paper, all of which can be instructive to future research endeavors in this area. First, future research would be well advised to incorporate other sources of data in the analysis. It is unclear

what influence the reliance by this analysis on the one source of data, albeit by various methods, had on the results. Data could be collected from other sources within the organization, asking top managers to describe the boundary spanning role played by the P/HR department. Additional data could be collected from other sources concerning objective characteristics of the organization's P/HR environment, such as the condition of the labor market in the area. A third source of additional data might describe the organization's profitability. Companies with little slack in their operating costs would be expected to have different concerns about their employees than those with discretionary funds. These examples point to the second limitation to this analysis. While each of the above additional data sources was considered in this analysis, the focus of the study on the Atlanta SMSA made their inclusion troublesome for the reason that variance in each was truncated. It is possible that the Atlanta area business environment is homogeneous enough as to not provide adequate variance on these measures, as well as those collected. Expansion of the present investigation to consider a broader, more heterogeneous, region would be desirable.

The third limitation is also a point which could be addressed by future research efforts. Ideally, the role of the boundary spanner in an organization should be tested longitudinally. Such a design would not only help to partial out the relationship between uncertainty and boundary spanning, but would also help understand the ability of the boundary spanner to influence organizational decision making. For example, it would be interesting to return for a "time 2" data collection to a company where the P/HR manager reported a great deal of concern about alcohol and drug abuse among employees at time 1 to see if the organization had used information gathered by the P/HR manager about an appropriate organizational response. Clearly this longitudinal design is more complicated, especially since the appropriate interval for a return visit is unclear, but in combination with

a more representative sample and additional data sources would provide a strong test of Thompson's theory.

There are two theoretical issues which were introduced in this paper that future research on boundary spanning might consider. The first is the concept of "consensual technologies" which was developed from institutionalization theories of organizations. This paper did not go further than to suggest that over time, organizations are able to reach a consensus on how to address environmental contingencies which had, at one time, presented the organization with uncertainty. Theoretically, this consensus is based in part on information provided to the organization by the boundary spanner, and is akin to the changes Thompson (1967) suggested organizations adopted to develop a structure that "fit" the environment. This concept is intimately tied to the second that is worthy of further consideration, that being the distinction between institutionalized and emerging environments. Of specific interest are the different strategies used by organizations to address contingencies in each. Future research, especially that incorporating a longitudinal design, could examine the process organization's use in developing such technologies and could potentially trace the relationship of an organization with an environmental sector as it became institutionalized.

In summary, this dissertation has made four contributions to the field. First, the literature review identified two weaknesses in the body of research employing Thompson's concept of boundary spanning. The suggestions that future research more carefully operationalize the concept and that attention be focused on the boundary spanning - uncertainty relationship both have important implications for future research. Second, the analysis was able to lend support to portions of Thompson's theory that had been largely untested. Third, the consideration of institutional and emerging environments as a way to frame the boundary spanning - uncertainty framework is argued to be a valuable conceptual

development and merits further consideration, as does the concept of consensual technologies. Finally, the use of Thompson's framework to describe the development of the P/HR function in organizations brings a rich set of hypotheses into an area of study that has to this point been largely atheoretical.

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Areas of interest include organizational theory, organizational behavior, personnel/human resource management, research methods and statistics, and employee assistance programs. Currently ABD, degree in Management with a minor in quantitative methods expected in June, 1989.

MA *Tulane University, 1984.* Degree in Applied Social Research with a concentration in research methods and data analysis.

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RESEARCH POSITION (since 4/84)

Research Associate to Paul M. Roman and Terry C. Blum. Duties include the following participation in a program of funded research while at Tulane University and Georgia Institute of Technology:

College of Management, Georgia Institute of Technology (6/87 to present).
Duties include a role in the direction and supervision of work on a five year grant (U.S. Public Health Service Grant No. AA-07218 from the National Institute on Alcohol Abuse and Alcoholism to Terry C. Blum, principal investigator) entitled "Worksite Integration of Alcohol/Drug/Mental Health Prevention Strategies". The grant is funded through 1992 and studies personnel/human resource practices and policies regarding issues such as alcohol and drug abuse, drug screening, health care cost containment, AIDS, and psychological troubles among employees through on site visits to 300 company locations in the Atlanta SMSA and throughout Georgia. Specific duties include instrument development, construction of the population frame and sampling, training of interview staff, supervision of the interview process, conducting on-site interviews, codebook development and data analysis.

Institute for Behavioral Research, University of Georgia (7/86 to 6/87). Duties included participation in the start up phase of a grant entitled "Medicalization: Substance Abuse Treatment and the Workplace" (U.S. Public Health Service Grant No. AA-01792 from the National Institute on Alcohol Abuse and Alcoholism to Paul M. Roman, principal investigator). The project is concerned with an examination of the medicalization of alcohol and other drug abuse in the workplace. The study focuses on substance abuse treatment centers as work organizations. Specific duties were concentrated on literature review, questionnaire development and conducting pre-test interviews.

Tulane University (4/84 to 7/86). Duties involved participation in all phases of a nationwide study entitled "The Structure and Content of Employee Assistance Programs" (U.S. Public Health Service Grant No. AA-01792 from the National Institute on Alcohol Abuse and Alcoholism to Paul M. Roman, principal investigator). Specifically, duties included questionnaire development and validation, sample construction, codebook development, coding and analysis. Specifically responsible for establishing and maintaining rapport with survey participants, as well as supervising all scheduling and personally conducting one hundred of the on site interviews.

PREVIOUS RESEARCH POSITION

Research Assistant, New Orleans Police Department (1/83 to 4/83). Developed and carried out a quantitative study of the victim-offender relationship in rapes and homicides which occurred in Orleans Parish in 1982. Consulted on departmental responses to study's findings.

PUBLICATIONS

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"Employee Assistance Programs: Prevention Concepts". Presented at the 1988 Meetings of the *American Public Health Association*, Boston, MA, November.

"An exploratory analysis of the reliability of the Lodahl and Kejner job involvement measure". *8th Annual I/O and O.B. Graduate Student Convention*. University of Tennessee, Knoxville, TN, April, 1987.

"The structure and content of internally based employee assistance programs". *Association of Labor-Management Administrators and Consultants on Alcoholism, National Meeting*, Boston, MA, November, 1986.

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"Qualitative comments regarding urban and rural settings for EAPs". *Association of Labor-Management Administrators and Consultants on Alcoholism, National Meeting*, Dallas, TX, November, 1985.

"External EAP service providers, regional and local comparisons". *Dallas Area Service Providers Meeting*, Dallas, TX, May, 1985.

"Employee assistance programs: an ethnographic assessment". *Southern Sociological Society*, Charlotte, NC, April, 1985.

"Structure and process in internal and external EAPs: A research update". *Texas EAP Symposium V*, Austin, TX, March, 1985.

RESEARCH IN PROGRESS

Bennett, N. "Boundary positions and boundary activity: A review and reconceptualization". First paper in partial fulfillment of the requirements for the Doctoral degree, College of Management, Georgia Institute of Technology.

Bennett, N. & Blum, T.C. "Employee attitudes concerning corporate drug testing practices: Implications for human resource management" (working title).

Bennett, N. & Milne, S. "Population ecology and open systems models of organizations: A contingency framework" (working title).

Blum, T.C. & Bennett, N. "Organizational decline and its consequences for human resource programs" (working title).

Bennett, N. & Stilwell, C.D. "An empirical examination of the Lodahl and Kejner job involvement measure" (working title).

PROFESSIONAL ACTIVITIES AND HONORS

Representative to the Academy of Management OB/OT Ph.D. Consortium, Los Angeles, August, 1988.

Member, Academy of Management.

Member, American Psychological Association.

Member, Southern Sociological Society.